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RESEARCH
PROGRAM ON
Wheat

CGIAR Research Program on Wheat
An Agri-Food Systems CGIAR
Research Program

Name of the CRP: **Agri-Food Systems CGIAR Research Program for Wheat**

Name of the Lead CGIAR Center: International Maize and Wheat Improvement Center (**CIMMYT**)

Flagship lead institutions (CGIAR Centers or lead partners)

Flagship 1: CIMMYT

Flagship 2: CIMMYT & ICARDA

Flagship 3: CIMMYT

Flagship 4: CIMMYT

Other participating CGIAR Centers: **ICARDA** – plus management partners **ACIAR (AU)** **BBSRC (UK)**, **ICAR (India)** – and platform partners **BARI (Bangladesh)**, **INIA (Bolivia)**, **JAAS (China)**, **EIAR (Ethiopia)**, **KALRO (Kenya)**, **INRA (Morocco)**, **IRESA (Tunisia)**, **INIAF (Uruguay)** – and international partners **G-20 Wheat Initiative**, **International Wheat Yield Partnership (iwyp.org)**. The complete list of partners is accessible [here](#).

[Acknowledging our funders:](#)

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EXECUTIVE SUMMARY

As MAIZE, RICE, PIM, and WHEAT joint foresight indicates, the cereal-based food security system has been evolving due to rural transformations. With an expected increase in population of 10.6 billion by 2050, the demand for cereals (including wheat) will rise to provide the base of global diets. The ability to supply more, and high-quality, food critically depends on the capacity from rural areas; thus, it is imperative to understand rural transformation. This study indicates that the cereal-based food system will remain dominant, so the challenges are to 1) support rural and agricultural transformation (e.g. poverty alleviation, employment creation) while maintaining agricultural diversity; and 2) for major cereals to meet the increased demand without displacing other nutritious crops and livestock from rural landscapes.

The CGIAR Research Program on Wheat (WHEAT) is well placed to provide the research and evidence for responding to these challenges. The CGIAR Phase II review of WHEAT concluded that the program has a track record of delivering local solutions with a global perspective and is well placed to continue on this trajectory in the coming decade.

WHEAT scientists and partners broke new ground in policy-informing climate change adaptation and mitigation research. WHEAT, CCAFS and national agricultural research system (NARS) partner scientists performed a meta-analysis comparing conservation agricultural practices with existing farmers' practice in South Asia. Zero tillage and residue retention offer 5.8% yield advantage and 25.9% economic advantage, with 12-33% less greenhouse gas (GHG) emissions compared to existing tillage-based practice. Similarly, as evidenced from comparisons in rice and wheat fields across the Indo-Gangetic Plains of India, Nutrient Expert-based fertilizer management lowered global warming potential by 12-20% in wheat systems, in addition to increased crop yield and incomes - see SLO 3.1, Table 1.

The global COVID-19 pandemic compelled scientists to be resilient, but also offered opportunities for supporting partners. WHEAT and MAIZE scientists, as part of a multi-Center initiative in Bangladesh, provided recommendations for policy action to the Secretary of Agriculture and Ministry line agencies. This has positioned the CGIAR as a "go-to" institution for advice on COVID-19 response in Bangladesh.

In order to fix our global agricultural system to prevent the next COVID-19, CIMMYT and partner scientists identified biodiversity stress and loss as a leading factor in the emergence of infectious diseases. Promoting "landscapes that work for nature and people" requires removing distorting policies and subsidies.

A WHEAT pathologist identified striking parallels between wheat rusts and human pandemics, and a WHEAT gender scientist surveyed male and female dryland farmers in Egypt and Tunisia about COVID-19 impacts and their coping mechanisms. MAIZE and WHEAT scientists engaged in three of four CGIAR COVID-19 Hub workgroups – see *OICR Table 3*.

COVID-19 has exacerbated rural labor shortages. Efforts in Latin America, sub-Saharan Africa and South Asia to expand appropriate agricultural mechanization solutions continued in 2020. Research revealed much higher demand for mechanization than previously found by macroeconomic analyses (Eastern and Southern Africa), pointing to a problem of access rather than demand – see *OICRs Table and section FP4*.

Part A: NARRATIVE SECTION

1. Key Results

1.1 Progress Towards SDGs and SLOs

Farmer impacts and uptake of conservation agriculture. WHEAT, CCAFS and NARS partners verified farmer benefits of adopting laser land levelling (LLL): on yields (rice +549 kg ha⁻¹; wheat +471 kg ha⁻¹) and net returns (an aggregate increase of US\$230/ha) in the rice-wheat production system of Haryana, India. Science-based evidence, enhanced capacity and systems optimization such as zero-tillage and adapted varieties led to advancing wheat seeding on more than 500,000 ha, benefiting over 800,000 smallholder farmers in the eastern Indo-Gangetic Plains – see SLO 1.1, Table 1.

In the Indo-Gangetic Plains of India, 0.87 million ha -- or 2.8% of total cultivated land -- is under zero-tillage (ZT), with multiple economic and environmental benefits. ZT led to significant farmer cost savings, equal to a 5% increase in average total household incomes. Nutrient Expert-based fertilizer management lowered global warming potential by 12-20% in wheat, compared to conventional fertilization, and increased crop yield and incomes – see SLO 1.2 & 3.2, Table 1.

Wheat seed scaling in Ethiopia. From 2015-2019, 131,000 rural households in Ethiopia directly benefited from improved seeds, with average yields among participating farmers increasing from 3.39 to 4.09 t/ha by 2018/19. Average yield of all rust-resistant varieties during the final crop cycle was 30% more than the yield of susceptible varieties (see p.94). Adoption intensity of rust resistant wheat varieties averaged 95% during 3 project years (p.92). A DNA-based study detailed that 10 varieties accounted for over 81% of the wheat area sampled; of these, 4 were recently released (post-2010) rust-resistant varieties -- corresponding to an estimated additional 225,500 tons of production in 2016/17, valued at US \$50 million. These achievements, if sustained, may boost the resilience for Ethiopian farmers in the face of increasing food insecurity during 2020 – See Table 1, SLO 1&2.

Scaling in Sub-Saharan Africa. TAAT-Wheat Compact reported major progress in Sudanese wheat production and productivity: With strong partnerships in place, private and public seed companies and farmer associations produced close to 65,000 tons of certified seed in 2019, compared to around 5000 tons in previous years. For the 2020/21 season, Sudan expanded its wheat area to 315,000 ha. At this rate, Sudan would reach self-sufficiency by 2024, according to Martin Fregene, Director for Agriculture and Agro-Industry, African Development Bank.

The Mexico-CIMMYT MasAgro partnership continued to manage 38 innovation platforms in 32 Mexican states (see pp.6-7), in which 17,000 farmers engaged, adopting 36 agronomic practices. By 2020, the 449th extension agent has graduated from the training program, which was set up in 2009. WHEAT scientists identified traits with genetic potential for Mexican wheat-growing areas, while 2 varieties obtained their commercial registration, driving quality seed production – see MELIA Table, S3069 & S3997.

1.2 CRP Progress towards Outputs and Outcomes

1.2.1 Overall CRP progress

WHEAT and partner scientists made progress on genetic gains, wheat blast, seed systems and agronomic practices (e.g. timely sowing) and further developed innovation pathways for scale-appropriate innovation. Both MAIZE and WHEAT, together with CCAFS, can point to significant new learnings on climate change mitigation and adaptation measures, as well as scaling – see OICR Table 3.

2020 Innovations. WHEAT and partner scientists generated new knowledge related to value chains (consumer demand, farmer adoption dynamics), climate smart agricultural practices (zero tillage, water management and precision fertilizer) and cereal-based nutrition benefits - as well as feeding the breeding products pipeline – see *Innovations Table 4*.

1.2.2 a) Progress by flagships

FP1: Enhancing WHEAT’s research for development (R4D) strategy for impact

Foresight about the greater potential impact of cereal staples showed that their full dietary contribution is often overlooked. Cereals are rich in bioactive components, such as dietary fiber, and have major health benefits. In the future, agriculture researchers should collaborate with other food systems stakeholders on nutrition-related challenges in cereal production, processing and manufacturing, food waste and losses. Cereal and food scientists could collaborate with social scientists to better understand the impacts on diets of the political economy of the food industry, and the diverse factors which influence local and global dietary transitions, including those based on novel and nutritious cereal-based products. Looking at urban realities today, a separate study found that most of the processed maize- and wheat-based foods sold in Mexico City were not healthy and such foods were associated with specific marketing techniques. Meanwhile, communications about WHEAT wholegrain research continued in 2020.

Scientists reviewed 47 adoption-impact studies in wheat agri-food systems in the Global South and found the coverage and scope did not fully capture the complex, dynamic technological changes taking place. Most studies only looked at short-term changes associated with innovation adoption for yield and profitability, often assuming that improvements in living standards of farm households follow de facto. Labor productivity, ecological and social impacts were largely ignored. The authors propose to enhance data availability and analytical rigor to reorient future research toward the less-explored but socially, economically, and ecologically relevant dimensions.

A scoping review of 54 studies found that satellite data has been successfully used to detect agricultural practices, but only a few studies have estimated yield impacts, despite being a fairly well developed approach. Additionally, few studies have explored impact estimation beyond the biophysical sphere. Estimation of certain environmental impacts of agricultural practices is possible through satellite data, but rarely done so far. Satellite data analysis allows information access with little delay, over longer periods of time and wide geographies. More interdisciplinary research is necessary to foster uptake of this alternative data source in R4D evaluations – see MEL Table.

Findings from a 2013 ICARDA-led survey of the Moroccan seed system document that, despite major national and international efforts, varietal turnover was very low. informed a 2019 “freedom to act” policy that resulted in the creation of over 300 new seed enterprises with more regional focus, enabling significant varietal turnover. – see *Policies Table*.

WHEAT scientists found that low adoption of new durum wheat varieties in Morocco was partially due to misalignment between breeders’ and farmers’ objectives. Interviews with 90 wheat breeders from 49 countries and a case study of 861 durum wheat growers in Morocco revealed the existence of five breeder and four farmer classes. North African breeders’ objectives did not group with any of the North African farmer classes, with breeders’ focus on disease resistance, while farmers were more concerned with yield stability and household transformation. This difference in breeder and farmer classes resulted in the selection of different genotypes to be promoted to varieties. The use of participatory farmer-weighted indexes could help raise farmer adoption levels.

FP2: Novel tools for improving genetic gains and breeding efficiency

A diversity analysis of 80,000 wheat accessions of CIMMYT and ICARDA’s internationally available collections; considered the largest in the world, highlighted that relatively little of the genetic

diversity available in landraces has been used in modern breeding. The clustering analysis identified landraces that have contributed to the genetic pool of modern breeding lines and those that offer unexplored alleles/diversity.

WHEAT scientists analyzed how the derivatives of “goat grass” (Aegilops tauschii) have been used to widen the genetic base for wheat breeding, through the development of synthetic hexaploid wheat (SHW). SHW is derived by crossing goat grass with durum wheat, serving as a bridge to transfer desirable traits into elite lines. Wheat scientists used 629 unique accessions from 15 countries for pre-breeding, producing 1577 primary SHWs. These derivatives represented 21% of the germplasm distributed by the CIMMYT genebank between 2000 and 2018, offering novel genetic diversity for high yield, heat and drought tolerance, and pest and disease resistance.

The Heat and Drought Wheat Improvement (HeDWIC) research team raised US\$ 5 million for a translational research and breeding Hub in Mexico and collaborated in 5 other proposals on heat tolerance leveraging the HeDWIC Hub. It also initiated collaborative research on a HeDWIC Elite Diversity Panel in Central/West Asia/North Africa. A Doctoral Training Program has been set up and discussions with the G20 Wheat Initiative’s AHEAD to expand the program are ongoing.

Researchers performed a comprehensive evaluation of genomic selection for 35 key traits in wheat evaluated by CIMMYT and national partners, and gathered strong evidence that Genomic Selection (GS) will be a very powerful tool for end-use quality related traits and resistance to some diseases in wheat. Genomic estimated breeding values for these traits have been routinely integrated into selection decisions. Integrating genomic breeding tools resulted in better selection efficiency and phenotyping cost-savings. GS can be beneficial for CIMMYT and other wheat breeding programs, particularly in developing countries that are constrained in their ability to evaluate a large number of breeding lines. Further information about progress in genomic prediction technologies under the Accelerating Genetic Gains in Maize and Wheat for Improved Livelihoods (AGG) project is available here (p.4).

FP3: Better varieties reach farmers faster

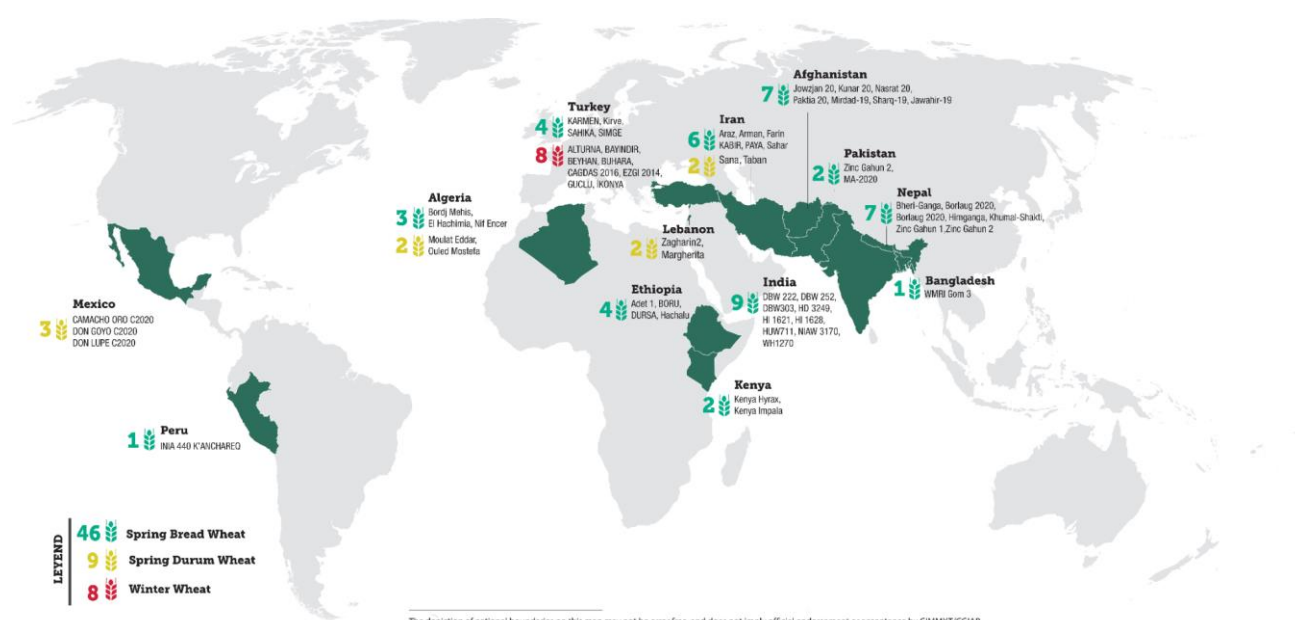
Update on rusts R4D. The frequent emergence of new stem rust virulent races remains a major constraint to wheat production. Surveillance results indicate *P. graminis* f. sp. *tritici* TTRTF, a race with complex virulence, is spreading rapidly and has become established in Ethiopia. Further spread is considered likely, and close monitoring is required. An overview of the current wheat rust situation is provided here. Further information about rusts research progress in the Accelerating Genetic Gains in Maize and Wheat for Improved Livelihoods (AGG) project is shown here (pp.6-8).

International germplasm exchange for research maintained during COVID-19 times. CIMMYT and ICARDA invest US\$700,000+ annually (mainly WHEAT W1&2) to distribute more than 350,000 packages of experimental seed (trials and nurseries) to over 220 IWIN collaborators worldwide. A majority of partners returned data voluntarily. Their key interest, according to a 2020 survey (87 respondents), was to obtain genetic diversity or parental germplasm (86.2%), or candidate varietal material (46%); most valuable to them was IWIN data on grain yield (76%) and diseases (75%). In addition, CIMMYT and ICARDA wheat genebanks fulfill seed requests from scientists globally on the basis of Standard Material Transfer Agreements (SMTA); this activity is funded by the Genebanks Platform – see Table 4.

WHEAT

CIMMYT distribution	2018	2019	2020
No. Recipient Countries	87	100	80
No. Recipients	423	580	392
No. Samples	300,464	149,919	166,396
No. Unique Accessions	101,758	26,038	37,870
No. Provider Countries	4	4	3
SMTAs CIMMYT	601	834	534
SMTAs ICARDA	81	56	76

At least 57 varieties were released in 12 countries in 2020, among them 7 new wheat varieties for multiplication and distribution to nearly 1 million wheat farmers and benefitting 28 million wheat consumers in Nepal. In Pakistan, long-term WPEP project (USDA) support resulted in the release of more than 40 high yielding, disease resistant varieties since 2010. See the table [here](#) and in Table 4.



Increasing genetic gain in research fields and on farmers' fields. Researchers estimated the genetic gains of the High Rainfall Trials (2006-2017) grown by NARS partners worldwide. Annual genetic gains were 3.8% compared with local checks (currently grown) of 1.17%, in the high-rainfall environment (160 and 65.1 kg ha⁻¹ yr⁻¹, respectively). In the low-rainfall environment, the genetic gains were 0.93% compared with local checks of 0.73 % (40 and 33.1 kg ha⁻¹ yr⁻¹) respectively. The annual Grain Yield (GY) genetic gain of the lines showed a significant phenotypic correlation between high- and low-rainfall environments in all examined years. The resulting genetic gains impact around 61 million hectares on which farmers grow CGIAR-derived varieties, on a 215 million ha total global wheat area – see *SLO Table 1, 2.1*.

Heat and drought resistant wheat in Pakistan (OICR 2019). During 2020, the CIMMYT-led Agricultural Innovation Program (*AIP-Wheat*) set up 10 new seed banks across the country for seed multiplication and farmer seed; four of these were in remote districts which face low wheat productivity and lack access to quality seed. Together with partners, AIP provided 10 tons of quality

seed to 200 member farmers (including 2 female farmers) and trained 107 farmers and agricultural extension staff.

An international collaborative effort developed a freely distributed Global Durum Wheat Panel (GDP) comprising 1,011 genotypes capturing 94–97% of the original diversity. This panel is distributed freely to 45 partners via the International Nursery system, and sets remain available to serve the need of more partners. Combining a massive community effort to phenotype this panel, with the available genotypic data (90K Illumina array), the GDP has empowered the wheat research community to discover novel and useful alleles and exchange them freely across international boundaries. As an [open resource available](#) to the whole community, the discovery of useful alleles can be immediately incorporated in breeding activities irrespective of the country or research group that makes the discovery. The panel was made possible financially by several international and national donors besides WHEAT: The Wheat Initiative, Norway/Crop Trust, USDA-ARS, GRDC and FAO.

WHEAT scientists found a clear superiority in crop wild relative (CWR)-derived lines in durum wheat breeding by examining 60 accessions from 9 countries and comparing top lines obtained via elite-by-elite crossing and CWR-derived lines planted in 8 different growing environments. Results revealed CWR-derived lines had superior grain size, as well as higher grain yield under low nitrogen and normal tillage.

Progress in wheat blast resistance. Sources resistant to wheat blast have been identified and are being used in breeding programs or candidates for varietal release. Bangladesh, Bolivia, India, and Nepal have released resistant varieties, and have conducted rapid seed production and dissemination. Resistance genes and sources of resistance are being used in CIMMYT and NARS partner breeding programs. Molecular markers identified can be used for marker-assisted selection on the global level. Wheat and partner scientists must urgently identify novel non-2NS sources of blast resistance; first successes were achieved. International trainings have strengthened the NARS capacity to mitigate the effects of wheat blast.

FP4: Sustainable intensification of wheat based-farming systems

Early sowing of wheat supports climate change adaptation. High-yielding wheat varieties with early heat stress tolerance enabled farmers of North India to advance sowing by two weeks. Early (October) sowing of wheat was augmented by machines that can plant under rice residue, avoiding burning. This package of new seed, machines and agronomy led to October sown wheat in >1 million hectares with about 0.8 t/ha extra yield and savings on irrigation. About ~0.3 million farmers benefitted from additional income (~US\$ 200/ha) and had reduced environmental footprints – see *SLO1.1, Table 1*

Crop production in the Middle East and North Africa (MENA) is characterized by increased monocropping, frequent soil tillage for weed control and delayed seeding, as well as by burning or on-site grazing of stubbles. Recent studies identified significant potential for Conservation Agriculture (CA) suitability and adoption by farmers, including for Morocco: 260,000 ha (33% of total cereal). However, no-tillage surface in Morocco, though doubled from 2013 values, stands at 21,000 ha in 2020 (Mrabet et al. 2021¹). MENA-wide, farmer CA practice covers less than 5% of total cereal area (2012 figures, see item 23), raising questions as to its suitability.

WHEAT and partner scientists showed that Conservation Agriculture-based practices have the potential to improve the economic and biophysical viability and sustainability of dryland agriculture in many MENA regions – see *MELIA Table*.

¹ Mrabet R, Moussadek R, Devkota M. and Rattan Lal. 2021. No-Tillage Farming in Maghreb Region: Enhancing agricultural productivity and sequestering carbon in soils. Chapter accepted. Book title. Soil Organic Carbon and Feeding the Future: Crop Yield and Nutritional Quality. In Advances in Soil Science Springer edition

Pilot analysis of robust nitrogen cycles in mid-hills and lowland (Terai) of Nepal. Nitrogen (N) is often the most limiting nutrient to productivity in smallholder mixed crop-livestock systems. The Terai farm N networks were organized (high productivity) but inflexible (poorly resilient) and consequently not sufficiently robust. Scenarios of improved management (improved seed, intercropping, use of fertilizers, better timing of activities) resulted in improved crop production, leading to reduced fodder imports and less N losses. N networks increased in flexibility, resulting in greater robustness of the N flow network in the farm systems. Increasing on-farm biomass production could be an important element on the way to sustainably intensify smallholder farms, especially when dependency on external resources can be reduced. WHEAT scientists consider a detailed analysis of nutrient flows and their robustness a suitable instrument for targeted improvement of nutrient use in smallholder crop-livestock systems.

Farm Power – Mechanization. Research has revealed much higher demand for mechanization in Eastern and Southern Africa, with African farming households being more dependent on labor markets than commonly assumed; and thus far more inclined to hire mechanization services. WHEAT and partner scientists compared case studies across 3 continents, reviewed Nepalese policy with a focus on appropriate-scale mechanization and investigated best-bet climate smart technologies, identifying the positive impact of appropriate mechanization on income and livelihoods; such as the Happy Seeder, which would offer farmers significant savings in production costs, amounting to 136 USD ha⁻¹, on top of saving water and making timely wheat sowing easier. Approximately 11,000 Happy Seeders are in use in north-west India; over 80% in Punjab. 18 manufacturers currently produce the Happy Seeder - see *OICRs Table*.

Landscape approaches. WHEAT and partner scientists contrasted forest restoration scenarios, to explore whether there are ideal configurations of forests that optimize benefits for food production. To do so, the researchers explored forest restoration scenarios in an agricultural-forest landscape in southern Ethiopia, where previous research showed increased wheat productivity in sections of fields adjacent to forests. Key findings include benefits to total landscape-level wheat production were almost twice as great when transitioning from low to medium- than when transitioning from medium-high to high tree cover. The benefits of hedgerows on agricultural production levelled off with increasing hedgerows. Considering a landscape perspective can help support food security goals and should play a more prominent role in forest conservation and restoration.

1.2.2 b) Relevance to COVID-19

Scientists were part of a multi-center initiative in Bangladesh and provided recommendations for policy action to the Secretary of Agriculture and Ministry line agencies. This positioned the CGIAR as a "go-to" institution for advice on COVID response in Bangladesh. With other partners, WHEAT and MAIZE scientists provided ex-ante, foresight and prioritization advice to decision-makers (see OICR 4025/MAIZE and 3988/WHEAT for details). With the goal of fixing our global agricultural system to prevent the next COVID-19, CIMMYT and partner scientists identified biodiversity stress/loss as a leading factor in the emergence of infectious diseases. Promoting landscapes that work for nature and people requires removing distorting policies and subsidies. Separately, ICRAF & CIMMYT scientists gathered evidence about value chain fractures and identified 3 priorities for further research. MAIZE and WHEAT scientists engaged in three of four COVID-19 CGIAR Hub working groups – see *Tables 3 and 10*.

CIMMYT and ICARDA implemented their Business Continuity Plan to protect the continuity of critical operations, steered by a Crisis Management Team that managed each centers' contingency plan. Monitoring tools and traffic light reports were also implemented to track project deliverables from all funding sources affected by COVID-19.

1.2.3 Variance from planned program for this year

No research lines were dropped. FPs' impact pathways and Theories of Change (e.g. direction) remain unchanged.

CIMMYT and ICARDA maintained their suspension of project operations and offices in Iran, because sanctions made even basic R4D operations impossible.

1.2.4 Altmetric score and publication highlights

The top three altmetric scores relate to peer-reviewed publications about germplasm improvement, with relatively many news mentions, and/or Twitter feeds and/or Mendeley readers: Multiple wheat genomes reveal global variation in modern breeding (500); Genetic variation for photosynthetic capacity and efficiency in spring wheat (109) & Diversity analysis of 80,000 wheat accessions reveals consequences and opportunities of selection footprints (97). A scoping review of adoption of climate-resilient crops by small-scale producers in low- and middle-income countries (75), based on a CERES2030 study relevant to both MAIZE and WHEAT, is a close fourth.

1.3 Cross-cutting dimensions (at CRP level)

1.3.1 Gender

Legacy impacts of GENNOVATE. Gender and social norms are among the main factors shaping decisions in the adoption of agricultural technologies. CGIAR and partner gender researchers have applied gender-responsive research strategies and methods to unravel the complex dynamics of these norms in agricultural settings around the world. Much of this research under WHEAT is shaped by the GENNOVATE (Enabling Gender Equality in Agricultural and Environmental Innovation) initiative (2014-2018), which engaged over 7,500 women and men from 137 agricultural communities in 26 countries across the Global South. This enabled discovery of new patterns, while maintaining emphasis on contextual specificity. Participating scientists pioneered a qualitative comparative methodology that program implementers could not have done on their own.

Evidence from GENNOVATE supported a powerful call for changing the way development researchers work to reach greater gender equity in agricultural innovations, with authors pointing out the need for an invigorated research agenda, which emphasizes critical self-reflection and introspection among research institutions about the norms they bring to the research process; partnerships with civil society and other organizations with long-term, trusted local presence; engagement with both women and men from different social groups on the structures and mindsets that hinder and enable equality and empowerment; sufficient time and resources to accompany a process of social change; and mechanisms to scale advances made using gender transformative approaches.

Recent studies. Researchers reported that when given similar access to production resources, women have a higher probability of adopting mini-tillers in Nepal. Other research highlighted that climate-smart agricultural technologies have the potential to reduce women's drudgery along with improvement in productivity and farm income. Research on gender norms and poverty dynamics from 32 villages in South Asia concluded that villagers associate movements out of poverty mainly with the actions and capacities of men, despite evidence revealing examples of women who exercise major roles in agriculture and actively work to improve the well-being of their families. Such evidenced experiences rarely alter normative beliefs and practices that entitle men to control women's and family resources.

WHEAT researchers proposed policy options to promote climate-smart technologies in India that have the potential to mitigate GHG emissions and reduce women's labor drudgery, with caveats about potential negative implications to some specific vulnerable populations.

Capacity Development. A Guidance Note was designed for researchers working in smallholder farming systems. It is useful to interventions planning community- and household-level action research with women and men farmers. The FAO-promoted compendium of 15 good practices of Gender Transformative Approaches (GTAs) includes individual templates for each practice, including the GENNOVATE as one of the best practices to contribute to GTA.

Outcomes. See OICR4032.

What have you learned? What are you doing differently? CGIAR and partner gender researchers have applied gender-responsive research strategies and methods to unravel the complex dynamics of gender and social norms prevalent in agricultural settings around the world. Gender and social norms are among the main factors shaping decisions in adoption of agricultural technologies.

Have any problems arisen in relation to gender issues or integrating gender into the CRP's research? With CRPs closing in 2021, there is little leeway to replace or commit new capacity/resources to gender research. Re-organization of the Gender Platform limited opportunities for joint actions, compared to 2017-19. Maintaining gender mainstreaming among CGIAR researchers requires additional efforts especially during this transition phase and into the future. The new CGIAR-GDI initiative may be helpful here.

1.3.2 Youth and other aspects of social inclusion / Leaving no one behind

Helping farmers "beyond reach" in Afghanistan. Wheat farmers in Afghanistan face challenges ranging from climate change, poor market support, limited mechanization and inadequate storage in the face of a demographic-driven rise in demand for wheat, which makes up around 60% of the nation's daily caloric intake. CIMMYT has been operating in the country since 2002, supporting national agricultural research systems to release high-yielding, disease resistant crop varieties. However, key elements for adoption include training, access to markets, and proximity to a main road. These are out of reach for many farmers, as anti-government forces continue to target transport infrastructure. A few recent developments offer hope. Four new high-yielding and disease resistant wheat varieties were released and the "Improving food security by enhancing wheat production and its resilience to climate change through maintaining the diversity of currently grown landraces" project successfully culminated in December 2019. With this project, researchers found that, on average, Afghan wheat landraces yielded highest under rainfed conditions when compared with those from Iran and Turkey, as well as against winter wheat trials carried out in 2018. These landraces have since been used in breeding improvement and crossing programs, as well as being multiplied and given back to local communities, making them more useful for those communities who grow them and conserving useful variability.

In Bangladesh, GENNOVATE researchers found that focusing on women in technical trainings and engaging women in social and professional networks, such as the Union Federation, allowed them to effectively express their agency, especially in the case of marginalized Santal and low-income Muslim women. A relatively simple empowerment process – training in wheat–maize technologies through a women's organization – has started to enable these women to secure access to innovations and start making important choices in their lives.

Food system recovery in fragile states. In support of FAO-led efforts to improve the Syrian seed security situation, ICARDA researchers in Aleppo and Homs helped rehabilitate the country's seed sector by producing basic seed stocks of wheat, barley, chickpea lentil. Seeds were multiplied and certified seed went to farming families, including 15,000 that planted one hectare of wheat (FAO

smallholder support program). ICARDA's experimental platform in the Bekaa Valley of Lebanon supports Syria's field experiments, seed multiplication, and capacity development for Syria's food systems.

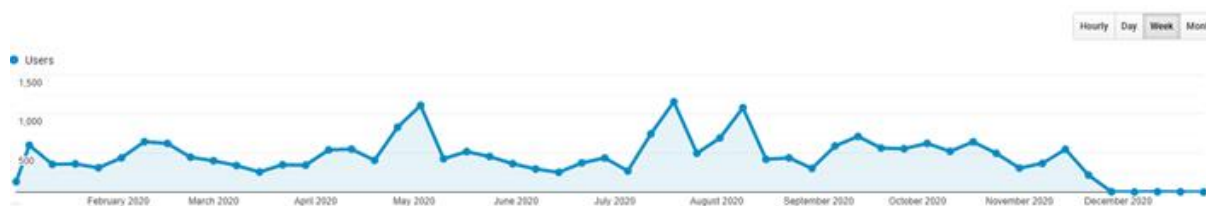
1.3.3 Capacity Development

See [also Table 8](#).

Safa Kumari, Head of ICARDA Seed Health, was recognized by the BBC as one of 100 women who are leaders of change and making a difference during these turbulent times. The 2020 Outstanding Paper Award of the Crop Science Journal by the Crop Science Society of America (CSSA) went to Dr. Khaoula El Hassouni's paper "Root system architecture and its association with yield under different water regimes in durum wheat." Hassouni is a former PhD student of ICARDA and University of Mohammed VI.

The 1st cohort of 22 Arab Women Leaders in Agriculture (AWLA) from 6 countries graduated in February 2020. WHEAT will also co-fund the AWLA (formerly Tamkeen) Program 2nd cohort, led by the International Center for Biosaline Agriculture (ICBA).

CIMMYT Academy was further improved: the system migrated to a new Learning Management System technology. Content curation and addition of new functionalities and connections to external e-learning providers (e.g. LinkedIn Learning, OpenSesame) will be 100% completed in 2021. CIMMYT Academy had 15,241 unique visitors, of which 43.5% are women (see graph below), which is up 26% from 2019. Current CIMMYT Academy users are CIMMYT staff and external users trained by CIMMYT scientists.



1.3.4 Climate Change

Are farmers willing to pay for climate-related traits of wheat? Based on a random sample of 303 smallholder farm households, WHEAT researchers established that **farmers are highly willing to pay for yellow rust- and frost-resistant traits**, nearly 10 times the value they are willing to pay for increased grain yield. They are also willing to **pay for harvesting wheat yield a month earlier**, at four times the value they attach to an additional 100 kg per hectare grain yield. This shows the importance farmers implicitly give to the climatic factors they must consider in their wheat production.

Conservation Agriculture for Sustainable Intensification (CASI) impacts in South Asia. Continued efforts of generating scientific evidence, innovative partnerships and enhanced capacity led to policy changes for large-scale adoption of low emission, risk-mitigating practices in smallholder systems. It has been estimated that adoption by South Asian farmers of about 2 million ha CASI and 20 million ha precision nutrient could potentially benefit ~14 million smallholders with ~14 million tons of extra grain production and US\$ 1.3 billion additional revenue generation, while reducing ~3.4 million tons of CO₂ emissions each year.

WHEAT scientists examined the prospects of smallholder production systems in South Asia (SA; Bangladesh, Bhutan, India, Nepal, Pakistan and Sri Lanka) in adapting to climatic variability while negative impacts on food systems. Researchers also investigated why farmers used few adaptation measures, if any. Adaptation is particularly fundamental to South Asian agriculture. The key

challenges are institutional change and flexibility to enable adaptation; the adaptive management approaches are new to the bureaucratic systems. To date, only a few studies have assessed the **overall effectiveness of adaptation actions taken at local, sub-national and national levels** - fewer still have focused on the **farmer behavior determinants of adaptation** to climate change. Researchers found that **education and interaction among farmers changed their adaptation behavior; focusing on farmer-to-farmer communication can help to improve adaptation** in agriculture. **Economic benefits alone may not explain farmers' adoption. Multiple factors, such as the gender of the household head, education and market access** affect adaptation behavior. Policies focusing on adaptation require the flexibility to address these factors adequately. In sum, **though adaptation practices are available to farmers, the institutional set-up to share them widely needs to be strengthened**. Researchers recommend designing dynamic policies for long-term climate change adaptation in agriculture, rather than a mere focus on agricultural technology.

2. Effectiveness and Efficiency

2.1 Management and governance

[The WHEAT-Independent Steering Committee \(WHEAT-ISC\) reviewed](#) and updated its recommendations about Must-Do WHEAT R4D during 2022 to 2030. It held a meeting together with MAIZE-ISC, to understand and discuss CRP closure matters, OneCGIAR consequences for WHEAT and MAIZE CRPs, as well as the future of independent scientific advice at the research program level under OneCGIAR.

Together with RICE, RTB and MAIZE-Independent Steering Committees, the WHEAT-ISC Chair undertook several efforts to put the future of independent, scientific/expert advice on the System Board's and Executive Management Team's OneCGIAR agenda, by recommending four advisory roles and joint feedback from an Agri-Food Systems perspective to the 2030 Research Strategy.

[The WHEAT-Management Committee managed](#) in-year financial buffers in response to changing SMO guidance about at-risk W1&2 amounts (e.g. DFID-UK contribution; USAID W2 transformed into W3 bilateral contribution to AGG project); dealt with COVID-19 consequences on both CGIAR and partner research delivery; pushed, largely unsuccessfully, for greater and earlier clarity on CRP closure and transition into 2022+ Initiatives - and directed remaining partner grant funds to closing research gaps (wheat blast in Zambia; high-BNI wheat) and documenting CRP impacts (e.g. DNA fingerprinting; assessment tools for sustainable intensification impacts).

Communications: 26,504 users visited the WHEAT Website, with 33,348 sessions counted; WHEAT Facebook experienced a 33% increase in followers, reaching 5,055 followers by the end of the year.

2.2 Partnerships

2.2.1. Highlights of External Partnerships

See also [Table 9](#) & Common Reporting Indicator on External Partnerships #C2.

[Cereal killer detective collaboration with John Innes Center \(UK\) and EIAR \(Ethiopia\)](#). The portable device [MARPLE \(Mobile and Real-time PLant disEase\) detects](#) which strain of the fungus pathogen is infecting a wheat crop in just 48 hours, so users can make fast decisions to control the disease and protect harvests. [MARPLE has been rolled out across 5 research hubs in Ethiopia](#), helping to prevent new pathogen strains entering from Asia, accompanied by [capacity development efforts](#) organized by the Alliance for Accelerated Crop Improvement in Africa (ACACIA).

In Nepal, it takes at least a year to collate the demand and supply of a required type and quantity of seed. A new digital seed information system is likely to change that, as it will enable all value chain actors to access information on seed demand and supply in real time. The information system is currently under development, as part of the Nepal Seed and Fertilizer (NSAF) project, funded by the United States Agency for International Development (USAID) and led by the International Maize and Wheat Improvement Center (CIMMYT), the Seed Quality Control Center (SQCC), Nepal Agriculture Research Council (NARC), the Centre for Crop Development and Agro Biodiversity Conservation (CCDABC) and the Vegetable Development Directorate (VDD).

WHEAT has been a partner to the [G20 Wheat Initiative](#) since 2012. This year, the initiative published [position papers on global data-sharing](#), setting up an informational infrastructure, and how to [build a GxExM systems agronomy framework](#) for wheat-based systems. WHEAT (CIMMYT) scientists continue to chair or co-chair 4 of the 11 Expert Working Groups.

2.2.2. Cross-CGIAR Partnerships (Centers, other CRPs and Platforms)

Collaborations with A4NH, Big Data, CCAFS, MAIZE, PIM and RICE were maintained. Discussions with WLE and FTA continued and will culminate in OneCGIAR Initiative proposals.

MAIZE, RICE, PIM, and WHEAT. The cereal-based food security system has been evolving due to rural transformations. With the expected increase in population to 10.6 billion by 2050, the total demand for cereals will rise, to provide for the caloric base of diets worldwide. IMPACT modelling pointed to these trends: Rural transformation in Asia is diversifying out of rice production, into high value commodities and non-farm employment. In Africa, smallholder rice production has growth potential as part of the rural transformation process. Rural transformation in MENA will certainly result in more pressure on the wheat sector, but investments in productivity enhancement could result in a significant increase of wheat supply. More value chain opportunities would also be expected with growing urbanization, markets and diversification. In Africa, besides rural transformation factors, the production and supply of major crops such as maize is influenced by environmental factors. Rain-fed maize systems in Africa will be hit hardest by climate change. To off-set these pressures, with rural transformation ongoing, investments are needed that allow farmers to take advantage of unexploited agricultural potential in maize-based systems.

2.3. Intellectual Assets

Have any intellectual assets been strategically managed by the CRP (together with the relevant Center) this year? Note: Further information can be [found here](#).

CIMMYT and ICARDA maintained their relevant policies to support greater transparency, including in partnerships and results dissemination, in line with the CGIAR Principles.

WHEAT is not a legal entity. The management of legal assets relevant to the CRP is managed by WHEAT Lead and participant centers, both of which annually submit a detailed and confidential intellectual asset report to the System Board and the information contained therein is not repeated here. Flagship projects, WHEAT and CIMMYT-PMU do not, on their own, manage intellectual assets without consulting with the Lead Center's legal department; the same applies to ICARDA.

(b) If relevant, indicate any published patents and/or plant variety right applications (or equivalent) associated with intellectual assets developed in the CRP and filed by Centers and/or partners involved in the CRP, giving a name or number or link to identify them.

CIMMYT and ICARDA have not filed, nor has any CIMMYT or ICARDA partner informed CIMMYT/ICARDA, of any application for patent or plant variety protection associated with intellectual assets (IA) developed under WHEAT.

(c) List any critical issues or challenges encountered in the management of intellectual assets in the context of the CRP (or put N/A).]

Critical issues and challenges that remain relevant:

- Ensure sufficient funding and adequate human resources to implement on a timely basis all actions needed for a proper IA management. During 2020, COVID-19 made replacing of specialist legal staff difficult.
- Lack of IP policies in some NARS; lack of knowledge among NARS of IA management practices at CGIAR Centers and/or insufficient capacity to conduct adequate IA management; in certain CGIAR donor country NARS, reluctance to apply ITPGRFA standards (STMA).
- Collecting, exporting and licensing seed in view of the ITPGRFA and the Nagoya Protocol.
- Some intellectual property policies or practices from certain MAIZE/WHEAT partners are not aligned with CGIAR IA management Policies.
- Harmonization of licensing practices to disseminate digital sequence data with the Open Access obligation, in light of concerns raised among some ITPGRFA stakeholders in relation to the use of such datasets.
- The rising bar for Centers' privacy protection and accountability in the context of dealing with datasets, wherein such data include personal information that carry with them accompanying dissemination obligations under Open Access.

2.4 Monitoring, Evaluation, Impact Assessment and Learning (MELIA)

This years' 28 foresight, ex-ante, and evaluation and learning products cut across the WHEAT Flagships. With a view to OneCGIAR and CGIAR Initiatives, several MELIA studies focus on breeding research innovations (3067) and improved germplasm-based impact assessment (3961); others on farmer/social adoption dynamics, looking at climate-smart agriculture (3964) or seed systems (3987). One study proposes an alternative to CGIAR country/regional collaboration approaches. A case is made for the continued critical role of cereals-legumes-based systems to reach SDG1 and SDG2. Cutting down on pollution and GHG emissions in India makes economic sense for farmers (3966).

WHEAT participated in the Monitoring, Evaluation and Learning Community of Practice (MEL CoP) Steering Committee for another year. It contributed to advancing several methodology inquiries related to the CGIAR annual reporting process. It also continued to enhance MARLO positioning the platform as a reporting tool for the upcoming One CGIAR Initiatives.

WHEAT also co-hosted the first virtual MEL CoP meeting replacing the CoP's annual face-to-face meeting given COVID-19.

WHEAT's lead Center continued building project management capacity, collecting data from previous CRP evaluations and converting them into institutional knowledge applicable to all projects. Collected evaluations, impact assessments, mid-term reviews, adoption studies and synthesis studies were stored in CIMMYT Project Management databases and translated relevant recommendations into project management best practices.

WHEAT was part of the CGIAR light touch reviews in 2020 to assess the effectiveness and the quality of science of the CRPs. Furthermore, WHEAT's lead center supported the WHEAT review conducted

by CGIAR Advisory Services. A series of recommendations and learnings resulting from the review were sent to CGIAR Science Leaders to shape the development of CGIAR Initiatives.

2.5 Efficiency

For improved efficiencies related to breeding research, see reporting under FP2 and FP3. CRP administration and management achieved no further efficiency gains since 2019.

2.6 Management of Risks to Your CRP

The three major risks remain unchanged during Phase II: W1&W2 budget insecurity and delayed transfer of W1&2 funds, which directly affects CRP research and development operations. WHEAT-MC and WHEAT-FP Leaders swiftly dealt with USAID's in-year decision to transform its W2 contribution into W3 co-funding of the bilateral AGG project. Agreed W1&2-funded research activities, which did not fit under AGG, were maintained by deploying in-year financial buffers.

Unfulfilled obligations by the partners for commissioned and competitive (sub)-grants; Due to COVID-19, several partners were granted no-cost extensions. WHEAT-MC continued to aim at closing of all partner grants by end September 2021.

Lack of a systematic and integrated approach for monitoring and evaluation at the outcome level. The [CGIAR results dashboard](#) reflects a more systematic MEL approach across the CRPs. CGIAR-System Office (SMO) led a quality assurance initiative of annual reporting, with a focus on outcomes and impacts reporting, as well as proper evidencing of achieved deliverables. As part of this Annual Report, WHEAT featured a greater number of Outcome and Impact Case Reports and reported new evidence under 4 SLO targets.

2.7 Use of W1-2 Funding

WHEAT is guided by the high-level framework for W1&2 deployment shown below, while Table 12 shows in more detail where W1&2 has been invested during 2020, based on [the 83 work packages in the W1&2-per-FP annual work plan](#) delivered CIMMYT and ICARDA scientists and their implementation partners.

	Strategic, longer-term research, seed invests	Rapid response (incl flexibility)	Cross-Portfolio, -CRP learning for impact	CRP Gov. & Mgmt.
Discovery (upstream)	FP1, 4: ex ante IA & ex post IA / adoption studies for new knowledge for better targeting, prioritizing; ARI, national partners FP2-4: Generate new knowledge for R-to-D pipeline: New alleles for heat and drought, other climate change-related traits identified; GS models using high throughput phenotyping and / or environmental data; Biological Nitrification Inhibition	FP3 new diseases & pests: Wheat blast in S-Asia	FP2-3: Germplasm improvement methodologies, methods, data mgmt (e.g. Genetic gain, cross-crops) FP4: Research on scaling out, innovation pathways	WHEAT-ISC, WHEAT-MC. SMB Member (DG), CRPs Rep in SMB, MEL CoP co-leadership
Validation	FP3: New traits into elite lines: Heat and Drought. Precision Phenotyping Platforms with NARS partners; expanded yield testing		FP4: Country coordination, systems research approaches	
Scaling out (down-stream)	FP1, FP4: Research on adoption dynamics, scaling out, targeting, prioritizing, M&E approaches	FP3-4: post-conflict	FP3.7, 4.4: Country coordination, companion crops into wheat-based	

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	FP3: Research on farmer adoption, seed systems innovation	emergency support	systems, capacity development	
CGIAR-SRF Cross-cutting themes	Gender / social inclusion applied to WHEAT innovations and rapid value chain assessments with proper gender lens		FP1, 4: AFS-CRPs & CCAFS FP3: WHEAT & A4NH on improved nutrition Inter-CRP: How to improve gender mainstreaming into research	

3. Financial Summary

In [2020, WHEAT received US\\$3.97 million W2](#) support from Australia (ACIAR) and UK (DFID) and \$6.5M W1 from Australia, Belgium, Canada, France, India, Japan, Korea, Netherlands, New Zealand, Norway, Sweden, Switzerland, UK and the World Bank. Total new W1&2 income was \$10.48 million, down from \$12.5 million (2019) and \$13 million (2018); however, \$1.96 million W2 was transformed into W3 linked to a specific bilateral project. Bilateral funder support is documented in the WHEAT Annual Financial Report and in the [CGIAR Financial Dashboards](#).

During 2020, WHEAT-MC

- Agreed to reduce the in-year buffer and spend on prioritized partner grants and CIMMYT/ICARDA-led work packages by September, based on updated figures from SMO.
- Committed over \$3 million, including some carry-over from 2019, to ongoing and new WHEAT partner grants; due to W1&2 uncertainty and COVID-19 implications just over \$2 million had to be carried over into 2021.

Part B. TABLES

Table 1: Evidence on Progress towards SRF targets

(Sphere of interest)

SLO Target (2022) Brief summary of new evidence of CGIAR contribution	Expected add'l contribution before end 2022	Geographic Scope
100 million more farm households have adopted improved varieties, breeds, trees, and/or improved management practices.	NA	•Geographic Scope: National. •Countries: Ethiopia.
	NA	•Geographic Scope: National. •Countries: Ethiopia.
	NA	•Geographic Scope: National,

	varieties led to advancing wheat seeding on > 500,000 ha, benefiting (>US\$200/ha) over 800,000 smallholder farmers in the eastern Indo-Gangetic Plains. Despite the increasing climatic risks, during 2018-19 the average grain yields in Bihar and eastern Uttar Pradesh have increased by 1.0 tons/ha, compared to base year (2009-10), with current yields of 3.0 and 3.7 tons/ha (Bihar/Eastern Uttar Pradesh). This translates to annual wheat production gains of ~0.5 million tons worth US\$ 145 million per year.		Regional. •Regions: Southern Asia. •Countries: India.
30 million people, of which 50% are women, assisted to exit poverty	DNA-based study documents the first nationally representative, large-scale wheat DNA fingerprinting study in Ethiopia (correctly identified 94% of the genotyped samples/plots) and reveals that new, rust-resistant bread wheat varieties were widely adopted. Germplasm originating from CGIAR centers has made a significant contribution. Corresponding productivity gains and economic benefits have been substantial, indicating high returns to investments in wheat improvement: Economic benefit estimates put the gain to farmers at additional 225,500 ton of extra production in 2016/17, valued at US\$ 50 M.	NA	•Geographic Scope: National. •Countries: Ethiopia.
	Ex post, CCAFS/WHEAT: Assessment of factors affecting the adoption of laser land leveling (LLL) and its impact on crop yields and net returns, based on HH survey data (621 randomly selected farmers, Haryana, India) and models. Adoption of LLL has significant positive impacts on yields (rice +549 kg ha ⁻¹ ; wheat +471 kg ha ⁻¹) and net returns (aggregate increase of US\$230/ha) in the rice-wheat production system, raising farmers' income substantially. LLL adoption at farm level is influenced by land size and quality, tenure system, availability of farm machinery (tractor), access to finance and farm cooperatives, gender of household head, level of education and training and access to extension services. These factors play a role in LLL scaling strategies aiming to generate large social, economic, and environmental benefits. There are 2.5M (wheat) farming households in Haryana, delivering 13.4% of Indian wheat production (2010-11). LLL adoption rates vary according to study focus.	NA	•Geographic Scope: National. •Countries: India.
	Ex post. Prevailing zero-till (ZT) practices in Bihar, India (ca. 77% of population, 104M (2021), engaged in agriculture; 10.5M wheat farming HH), led to significant yield gains in 3 out of 4 years, notably in less favorable seasons (unique panel dataset, 961 farm households in 87 villages). Estimated yield increase was 660 kg ha ⁻¹ . ZT led to significant farmer cost savings in all four seasons, equal to a 5% increase in average total household income. Policy-makers in Bihar and adjacent states should continue to support expansion of ZT service providers networks;	NA	•Geographic Scope: National. •Countries: India.

	farmers might achieve even greater benefits from fully adopting Conservation Agriculture practices. Degree of land fragmentation may influence ZT adoption. The average size of ZT adopters' largest plot consistently exceed that of non-adopters. Data shows that productivity impacts of early sowing vary across agro-ecological zones; and that genotype x management interactions affect wheat productivity under ZT versus conventional practices.		
Improve the rate of yield increase for major food staples from current < 1% to 1.2-1.5% per year	Ex post (modelling), Grain yield progress over 50 years of spring wheat breeding at CIMMYT was determined in field trials conducted 2013–2017. Trials included 30 varieties (24 bread wheat and 6 durum wheat) released between 1965–2014 and were sown under managed optimum, drought and heat stress conditions. Grain yield progress per year for bread wheat ranged from 31.2 kg/ha, 35.3 kg/ha, and 24.7 kg/ha in optimum irrigated environments. In 2 different drought and one heat-stressed environments, bread wheat grain yield progress was estimated as 25.6 kg/ha, 17.7 kg/ha, and 18.1 kg/ha per year; similar results arose for durum wheat. Researchers highlight a continuous upward trend in the breeding progress of the CIMMYT/CGIAR wheat breeding program in a range of environments and management conditions. The study points to drip irrigation systems that make judicious use of water resources, while increasing yield potential. This study estimated the genetic gains of the High Rainfall Trials (2006-2017) grown by NARS partners worldwide. In high-rainfall environments, annual genetic gains were 3.8% and compared with local checks, 1.17 % (e.g. additional yield of 160 and 65.1 kg/ha/yr, respectively). In the low-rainfall environment, genetic gains were 0.93 and 0.73 % (40 and 33.1 kg/ha/yr), respectively. Several of the five best performing lines were common in both environments. The results indicate genetic gain ranging from 0.5-1% per annum, which on average represents 46.6 kg/ha/year. The above described genetic gains impact ca. 61 million ha on which farmers grow CGIAR-derived varieties, of 215 million ha of the global wheat area. Without the exotic (foreign) germplasms from CIMMYT and other countries, the Chinese average annual yield would have been lowered by 63 kg/ha and 323 kg/ha, respectively, during 1982–2014. CGIAR (CIMMYT) germplasms' contribution to annual yield increased from 11 kg/ha (1982/85) to 134 kg/ha in 2011/14, increasing by more than 11 times. When compared	NA	•Geographic Scope: Global, Regional. •Regions: Central Asia, Eastern Asia, Latin America & the Caribbean, Northern Africa, Southern Asia, SSA, W Asia.
		NA	•Geographic Scope: Global, Regional. •Regions: See row above
		NA	•Geographic Scope: National. •Countries: China.

with Chinese germplasms-alone varieties, on average a 1%-point increase in the share of the varieties with exotic germplasms alone (incl. CIMMYT) raised wheat yields by nearly 0.2%, and to nearly 0.3% for varieties with both Chinese and exotic germplasms.

The **estimated increase in wheat production due to exotic germplasms** reached 10.4 million tons **annually** and has increased over time in the past three decades. The accumulated contributions of CIMMYT's and Others' germplasms to Chinese wheat production amounted to 56 million tons (1.7*33 years) and 287 million tons (8.7*33), respectively, for the period 1982-2014.

30 million more people, of which 50% are women, meeting minimum dietary energy requirements

150 million more people, of which 50% are women, without deficiencies of one or more of the following essential micronutrients: iron, zinc, iodine, vitamin A, folate, and vitamin B12

10% reduction in women of reproductive age who are consuming less than the adequate number of food groups

5% increase in water and nutrient (inorganic, biological) use efficiency in agro-

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ecosystems, including through recycling and reuse

Reduce agriculturally-related greenhouse gas emissions by 0.2 Gt CO₂-e yr⁻¹ (5%) compared with business-as-usual scenario in 2022

55 million hectares (ha) degraded land area restored

2.5 million ha of forest saved from deforestation

Ex post, WHEAT, CCAFS: Conservation Agriculture for Sustainable Intensification (CASI) adoption studies are summarized under 1.1, 1.2 & 2.1 above. A meta-analysis of 9686 paired comparisons ^{NA} between conservation agricultural practice and existing farmers' practice in South Asia showed that zero tillage and residue retention offer 5.8% yield advantage and 25.9% economic advantage, with [12-33% less GHG emissions](#) compared to existing tillage-based practice. Similarly, as evidenced from comparisons in rice and wheat fields across Indo-Gangetic Plains of India, Nutrient Expert-based fertilizer management [lowered global warming potential by 12-20% in wheat](#), in addition to increased crop yield and incomes. The bottom-up analyses in India and Bangladesh suggest that zero tillage & residue management, as well as precision nutrient management, [could reduce country-level GHG emission by 17 and 17.5 million tons CO₂ emissions per year in India](#) ; and by 0.55 and 7.98 million tons CO₂ per year in Bangladesh.

- Geographic Scope: Regional.
- Regions: Southern Asia.

Table 2: Condensed list of policy contributions in this reporting year

(Sphere of Influence)

Title of policy, legal instrument, investment or curriculum to which CGIAR contributed (max 30 words)	Description of policy, legal instrument, investment or curriculum to which CGIAR contributed (30 words). See guidance for what to cover.	Level of Maturity	Link to sub-IDOs (max. 2)	CGIAR cross-cutting marker score				Link to OICR or link to evidence
				Gender	Youth	Capdev	Climate Change	
651 - Conservation Agriculture for Sustainable Intensification (CASI) in Eastern India	Policy brief by ICAR, CIMMYT & partners lays out clear case to policy-makers about benefits of CASI plus a road map for accelerating its adoption in Eastern India.	Level 1	• Agricultural systems diversified and intensified in ways that protect soils and water • Closed yield gaps through improved agronomic and animal husbandry practices	1 - Significant	1 - Significant	2 - Principal	1 - Significant	OICR2764
659 - Gather evidence on soil and root health challenges in 5 CWANA countries to initiate policy action	Soil Borne Diseases at CIMMYT focuses on diseases attaching to root systems. Healthy roots store more CO2 in the soil and reduces the negative effect of CC. Several surveys conducted in Algeria, Morocco, Kazakhstan, Azerbaijan and Kyrgyzstan in last 2 years. Of significant value to policy makers is to understand the harm of the hidden enemies. Ultimately, partners will take the evidence to decision makers, to consider	Level 1	• Enhanced institutional capacity of partner research organizations • Improved forecasting of impacts of climate change and targeted technology development • Reduced biological and chemical hazards in the food system	1 - Significant	0 - Not Targeted	1 - Significant	1 - Significant	

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	implementing strategies to better contribute to soil and root health management. Soil Borne Pathogens Program at CIMMYT partnered with Syngenta to implement a Forum on Soil and Root Health, with more than 800 participants, to raise greater awareness in CWANA & Africa.							
682 - Improve Nepalese mechanization policy: Focus on appropriate-scale and gender equality	WHEAT/MAIZE scientists recommend 3 strategies to promote responsible agricultural mechanization, and that policymakers implement female-targeted farm mechanization programs to enhance the level of small-scale mechanization in Nepal's hilly regions.	Level 1	• Closed yield gaps through improved agronomic and animal husbandry practices	1 - Significant	0 - Not Targeted	0 - Not Targeted	1 - Significant	OICR2764
686 - Freedom to act in Moroccan seed sector improves wheat varietal replacement	WHEAT seed system research was a significant input to Morocco Gov revamping of the national seed business; 300 new seed businesses established.	Level 2	• Adoption of CGIAR materials with enhanced genetic gains	0 - Not Targeted	0 - Not Targeted	0 - Not Targeted	0 - Not Targeted	OICR3994
694 - Faster access to improved varieties for smallholder farmers growing wheat on 40 million ha in South Asian countries	Breeding-to-variety release cycle cut by 2 years in India thanks to phenotyping and meta-GWAS for grain yield, heat tolerance, resistance to yellow rust and spot blotch.	Level 1	• Adoption of CGIAR materials with enhanced genetic gains	0 - Not Targeted	0 - Not Targeted	1 - Significant	0 - Not Targeted	OICR3285

Table 3: List of Outcome/ Impact Case Reports from this reporting year

(Sphere of Influence)

Title of Outcome/ Impact Case Report (OICR)	Link to full OICR Maturity level
OICR3929 - Early sowing of heat-tolerant wheat, new machinery and agronomic management enable 1.1M farmers in South Asia: Greater resilience, productivity and climate change adaptation	Link Level 3
OICR3930 - WHEAT and MAIZE science, partnerships, outreach lead to policies for conservation agriculture and precision nutrient management in South Asia, mitigating climate change, increasing smallholder income	Link Level 1
OICR3988 - Recommendations for policy action to the Secretary of Agriculture and Ministry agencies in Bangladesh, and policy advice to decision makers in India, Mexico and Nigeria	Link Level 1
OICR3994 - WHEAT/ICARDA analysis and outreach enables government policies that developed the Moroccan national seed sector, adding 300 new seed businesses	Link Level 2
OICR4020 - Three WHEAT/CIMMYT elite lines released by ICAR as varieties for North West India's smallholder farmers in record time	Link Level 1
OICR4030 - Sources of resistance to wheat blast and its underlying mechanism identified, resistant gene(s) introgressed and resistant varieties released in Bangladesh, India and Nepal	Link Level 1
OICR4031 - Significant genetic gains in grain yield delivered by breeding research: A constant flow of high-yielding, disease-resistant germplasm provided to National Agricultural Research System partners	Link Level 3
OICR4032 - Adoption and Scaling Interventions are progressively guided by gender-responsive research methods across CGIAR CRPs, including WHEAT	Link Level 1
OICR4042 - Cross-region/project learning from appropriate-scale mechanization projects: Rural labour dynamics, access and scaling approaches are key issues	Link Level 2

Table 4: Condensed list of innovations by stage for this reporting year

For IWIN distribution 2020, see page 9 in narrative section.

Title of innovation with link	Innovation Type	Stage of innovation	Geographic scope (with location)
1918 - Use of a positive deviance approach to inform farming systems redesign in Bihar, India	Research and Communication Methodologies and Tools	Stage 1: discovery/proof of concept (PC - end of research phase)	Regional, Southern Asia
1921 - New typology of women's strategies to strengthen their managerial decision-making power in wheat	Social Science	Stage 3: available/ ready for uptake (AV)	Global
1948 - Protocol for durum wheat milling into semolina	Research and Communication Methodologies and Tools	Stage 3: available/ ready for uptake (AV)	Global
1998 - Mid-density, low cost genotyping platform "DArTAG" for use in genomic selection of wheat.	Genetic (varieties and breeds)	Stage 2: successful piloting (PIL - end of piloting phase)	Global
2015 - Identification of new pleiotropic loci conferring resistance to Leaf Rust and Yellow Rust in wheat on chromosome 1B in "Arabelu" and "Mucuy" populations	Genetic (varieties and breeds)	Stage 2: successful piloting (PIL - end of piloting phase)	Global
2016 - Identification of new genomic regions conferring leaf rust resistance in durum wheat	Genetic (varieties and breeds)	Stage 2: successful piloting (PIL - end of piloting phase)	Global
2017 - Efficient molecular markers for screening wheat grain arabinoxylan content (dietary fiber) developed	Biophysical Research	Stage 3: available/ ready for uptake (AV)	Global
2023 - Breeding methods innovation: Competitive Allele-Specific PCR (polymerase chain reaction) markers associated with wheat phenotypic traits were developed.	Genetic (varieties and breeds)	Stage 1: discovery/proof of concept (PC - end of research phase)	Regional, Central America

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2025 - Machine learning innovations for rapid assessments in wheat and rice systems	Production systems and Management practices	Stage 1: discovery/proof of concept (PC - end of research phase)	Regional, Southern Asia
2049 - Wheat Yield Collaboration Yield Trial (WYCYT) nursery and International Wheat Yield Partnership Yield Potential Trait Experiment networks released in 2020	Genetic (varieties and breeds)	Stage 3: available/ ready for uptake (AV)	Global
2058 - Stripe rust associated genotyping-by-sequencing markers for marker-assisted selection and genomic profiles of 52,067 wheat breeding lines for stripe rust associated markers.	Biophysical Research	Stage 1: discovery/proof of concept (PC - end of research phase)	Multinacional, India, México, Kenya
2060 - Blast associated genotyping-by-sequencing markers for marker-assisted selection and genomic profiles of 4,143 wheat breeding lines for the 2NS translocation.	Genetic (varieties and breeds)	Stage 1: discovery/proof of concept (PC - end of research phase)	Multi-national, Bolivia (Plurinational State of), Bangladesh
2131 - Implications for applying El Niño Southern Oscillation (ENSO)-based precipitation outlooks took agricultural management decisions in Ethiopian Highlands	Biophysical Research	Stage 3: available/ ready for uptake (AV)	National, Ethiopia
2132 - Set of solutions using remote sensing and supervised learning to replace autoregressive integrated moving average models to forecast weather	Other	Stage 2: successful piloting (PIL - end of piloting phase)	Global
2133 - Support of policy makers in Bangladesh to address the COVID19 pandemic based on foresight analysis.	Social Science	Stage 3: available/ ready for uptake (AV)	National, Bangladesh

Table 5: Summary of status of Planned Outcomes and Milestones

(Sphere of Influence-Control)

FP	FP Outcomes 2022	Sub-IDOs	Summary narrative on progress against each FP outcome this year.	Milestone	2020 milestones status	Brief Explanation Provide evidence for completed milestones (refer back to means of verification, and link to evidence wherever possible) or explanation for extended, cancelled or changed	Link to evidence
FP FP1 Outcome: 1.8	1	National and regional policy makers improved policy-making and increased investment based on evidence	• Increase capacity of beneficiary ies to adopt research outputs Ex-ante/spatial assessments assess how major drivers will alter WHEAT and agri-rural landscapes in the Global South. Study highlights to inform policy: COVID-19: monitoring food system disruptions (2); Rapid Assessment Food-Nutrition Security Bangladesh (13); Climate-change analysis: adaptation options Ethiopia (1), South-Asia (4); mitigation options South-Asia (3); latitudinal--uncertainty modeling (12); Wheat-disease monitoring/decision-making tools: Bio-economic global assessment leaf-rust (6); first-report new stem-rust race (11); rust surveillance Ethiopia (14); Modeling: Multi-temporal yields (5); yield--phenology China (9); socio-economic factors (7); Outlook: Rural transformation and agri-food systems (8); Ethiopian agricultural extension (15); predicting adoption (10).	2020 - Climate change and other dynamics integral to foresight research, showing how they transform agri-rural landscapes.	Complete	Foresight/targeting 15 publications: 12 papers, 1 report, 1 book chapter, 1 brief: 1. Weather & Climate Extremes 29, 100263. 2. Food Security 12, 761-768 3. Environment, Development and Sustainability 22, 3267–3289. 4. Environment, Development and Sustainability 22, 5045–5075. 5. Precision Agriculture 21, 1263-1290. 6. Phytopathology 110, 1886-1896. 7. Crop Science 60, 568-5811. 8. Global Food Security 26, 100441. 9. Ecological Modelling 430, 109132. 10. Applied Economic Perspectives & Policy 42, 100-112. 11. Plant Disease 104, 293-293. 12. Nature Food 1, 63-69. 13. Report – 2nd Rapid Assessment of Food and Nutrition Security in the Context of COVID-19 in Bangladesh. FAO - IFPRI - IFAD - CIMMYT - IRRI - WorldFish, Dhaka. 14. Book-chapter: Enhancement of rust surveillance, early warnings and phenotyping of wheat genotypes in Ethiopia 15. Brief - Ethiopian Agricultural Extension - Where to go?	1. https://doi.org/10.1016/j.wace.2020.100263; 2. https://doi.org/10.1007/s12571-020-01083-2 3. https://doi.org/10.1007/s10668-019-00345-0 4. https://doi.org/10.1007/s10668-019-00414-4 5. https://doi.org/10.1007/s11119-020-09719-1 6. https://doi.org/10.1094/PHYTO-02-20-0032-R 7. https://doi.org/10.1002/csc2.20114 8. https://doi.org/10.1016/j.gfs.2020.100441 9. https://doi.org/10.1016/j.ecolmodel.2020.109132 10. https://doi.org/10.1002/aep.13012 11. https://doi.org/10.1094/PDIS-07-19-1390-PDN 12. https://doi.org/10.1038/s43016-019-0004-2 13. https://knowledgecenter.cimmyt.org/cgi-bin/koha/opac-detail.pl?biblionumber=62500 14. https://repository.cimmyt.org/handle/10883/20837 15. https://repo.mel.cgiar.org/handle/20.500.11766/10104

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FP1 Outcome: 1.10 Farmers have greater awareness and access to, and increased adoption and adaptation of improved technologies	Highlights to enhance adoption/impacts: - DNA fingerprinting wheat (Ethiopia): (7,8) - Adoption/impact reviews: wheat theory of change. studies (10); remote-sensing (11); NW Mexico (16) - Adapting to climate change: India (2,4), Pakistan (1) - Adoption/impact studies: India (3,9), Pakistan (12, 14), Nepal (13), Morocco (5), Syria (6), Ethiopia (15)	2020 - Adoption and impact studies on technologies- rolling plan based on progress of technologies along the	Complete	Adoption/impact 16 publications: 14 papers, 1 book chapter, 1 report: 1. Arabian Journal of Geosciences 13, 1025 2. International Journal of Climate Change Strategies and Management 12, 128-146 3. Natural Resources Forum 44, 255-273 4. International Journal of Innovation and Sustainable Development 14, 219-245 5. Agronomy 10, 998 6. Sustainability 12, 3640 7. Scientific Reports 10, 18532 8. PLOS ONE 15, e0235484 9. International Journal of Agricultural Sustainability 18, 567-593 10. Outlook on Agriculture 49, 89-98 11. Agronomy for Sustainable Development 40, 16 12. Journal of Animal and Plant Sciences 30, 175-184 13. Journal of Agricultural Economics 71, 199-218 14. Asian Development Review 37, 185-212 15. Book chapter - Impact of rust resistant wheat varieties on the productivity 16. Report - Historia de la transformación agrícola del Valle del Yaqui, Sonora. 1981-2013	1. https://doi.org/10.1007/s12517-020-06019-w 2. https://doi.org/10.1108/IJCCSM-09-2018-0065 3. https://doi.org/10.1111/1477-8947.12197 4. http://doi.org/10.1504/IJISD.2020.106243 5. https://doi.org/10.3390/agronomy10070998 6. https://doi.org/10.3390/su12093640 7. https://doi.org/10.1038/s41598-020-75181-8 8. https://doi.org/10.1371/journal.pone.0235484 9. https://doi.org/10.1080/14735903.2020.1794490 10. https://doi.org/10.1177/0030727020930728 11. https://doi.org/10.1007/s13593-020-0610-2 12. http://www.thejaps.org.pk/Volume/2020/30-01/20.php 13. https://doi.org/10.1111/1477-9552.12333 14. https://doi.org/10.1162/adev_a_00146 15. https://repository.cimmyt.org/handle/10883/20837 16. https://repository.cimmyt.org/
	Highlights to enhance gender/social-inclusiveness: - Country case gender-wheat studies: Ethiopia (19, 21); Nepal (20); Bangladesh (22) - Multi-country gender-wheat review studies: South Asia (24) - GENNOVATE knowledge products (18, 25, 26) - Quantitative gender studies: India (17), Nepal (23)	2020 - Assessment of gender influence on men/women's capacity to innovate in WHEAT AFS-based livelihoods.	Complete	Gender/social-inclusiveness 10 publications: 8 papers, 1 book chapter, 1 report 17. Review of Development Economics 24, 973-990 18. Development in Practice 30, 541-547 19. Sustainability 12, 9847 20. Sustainability 12, 374 21. Women's Studies International Forum 83, 102420 22. Canadian Journal of Development Studies 41, 20-39. 23. Technology in Society 61, 101250 24. International Journal of Community Well-Being 3, 289-310 25. Book chapter - GENNOVATE 26. Brief - GENNOVATE resources for scientists and research teams	17. https://doi.org/10.1111/rode.12670 18. https://doi.org/10.1080/09614524.2020.1757624 19. https://doi.org/10.3390/su12239847 20. https://doi.org/10.3390/su12010374 21. https://doi.org/10.1016/j.wsif.2020.102420 22. https://doi.org/10.1080/02255189.2019.1650332 23. https://doi.org/10.1016/j.techsoc.2020.101250 24. https://doi.org/10.1007/s42413-019-00047-5 25. https://doi.org/10.4060/cb1331en 26. https://repository.cimmyt.org/handle/10883/20934

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FP1 Outcome: 1.9 Last mile provider (extension of partners, farmers organizations, community-based organizations, private sector) increased access and promotion of technologies to farmers	• Increase capacity of beneficiaries to adopt research outputs	Market/value chain development refines the understanding of agri-food systems and develops a strategic agenda for WHEAT and included benchmarking information of wheat value chain collected in selected countries. Highlights of studies to enhance last mile linkages: - COVID-19: Assessing economic loss and food security options, Bangladesh (2) - Farmer interest/willingness-to-pay for traits: participatory farmers-weighted selection indices durum, Morocco (1); climate related traits, Ethiopia (4) - Wheat value chain development (Ethiopia): governance structures--incentives (3); integrating smallholders durum (6); agricultural extension (7). - Agri-nutrition research: assessing contribution to human nutrition--health (5 – published online, linked to visiting fellow)	2020 - Benchmarking information of wheat value chain collected in selected countries to support identified of priorities and effective interventions.	Complete	Mkt/VC 7 publications: 5 journal papers, 1 book chapter, 1 brief: 1. Crop Breeding, Genetics and Genomics 2, e200014 2. PLOS ONE 15, e0240709 3. African Journal of Agricultural and Resource Economics 15, 157-176 4. Agricultural Systems 185, 102947 5. Food Policy, 101976 6. Book chapter - Integration of smallholder farmers into the durum wheat value chain: Achievements, challenges and lessons learned 7. Brief - Ethiopian Agricultural Extension - Where to go?	1. https://doi.org/10.20900/cbagg20200014 2. https://doi.org/10.1371/journal.pone.0240709 3. http://afjare.org/volume-15-no-2/ 4. https://doi.org/10.1016/j.agsy.2020.102947 5. https://doi.org/10.1016/j.foodpol.2020.101976 6. https://repository.cimmyt.org/handle/10883/20837 7. https://repo.mel.cgiar.org/handle/20.500.11766/10104
FP FP2 Outcome: 2.4 2 Crop researchers world-wide increased use of novel germplasm and tools for validation, refinement and development of products	• Adoption of CGIAR materials with enhanced genetic gains	The continued development and distribution of pre-breeding germplasm to colleagues within the CGIAR and to NARES partners was a highlight of a challenging year. The synthesis of work on sparse testing provided clear guidelines on effective implementation of sparse testing, highlighting the opportunities and potential pitfalls of the approach. Challenges in transformation of wheat resulted in a slowing of progress in gene editing, newly-sourced resources offer potential	2020 extended to 2021 - Gene editing successfully implemented for Lr67 and/or MLO genes to develop tolerance to wheat diseases.	Extended	Transformation of elite wheat lines has proven to be challenging, thus has presented a bottleneck in editing genes. Although we have generated more than two dozen transgenic plants with the editing machinery for the Lr67 gene in Reedling, none carried an edited copy of the gene. We have recently acquired new vectors from UC Davis that have been shown to improve both transformation and gene editing efficiencies. These vectors containing guide RNA molecules for Lr67 and MLO genes have just been transformed into Reedling. We expect to see results in 5-6 months and hope to identify edited copies for both the genes.	https://www.dropbox.com/work/WHEAT%20FP2%20Editing%20evidence?preview=Lr67+and+MLO+-+CRP+Milestone+2020.docx

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to address this challenge in 2021. Finally, COVID-19 heavily impacted the initiation of doctoral training under HeDWIC, this important endeavor will extend into 2021.

2020 - Demonstrated use of genomic prediction to save phenotyping costs via 'sparse testing'.	Complete	Sparse testing implies modifying the original multi-environment breeding trials system (where all breeding germplasms are evaluated in all the testing environments) into a system where not all the breeding lines are sown in all environments. How best to establish a multi-environmental trial system that will be economically feasible, save land and water resources, without highly affecting the precision by which breeding lines or varieties' performance are assessed, predicted and selected? Sparse testing was applied to maize and wheat breeding of CIMMYT in order to study the genomic-enabled prediction accuracy of different field sparse testing systems. We studied three sparse testing allocation designs: (1) no-overlapping lines in environments, (2) complete overlapping (0 non-overlapping) lines tested in all of the environments and (3) combinations of the two previous cases where certain number of non-overlapping/overlapping lines were distributed in the environments. The study used three extensive wheat data sets and two large maize data sets. Results for sparse testing on wheat and maize data sets indicated that substantial savings on testing resources can be achieved by allocation designs and genome-based models. Reducing the size of the calibration sets always reduced the genomic-enabled prediction accuracy. The study manuscript is currently under peer review.	https://data.cimmyt.org/dataset.xhtml?persistentId=hdl:11529/10548552
2020 - Phenotypic, genotypic and genealogical data from the previous three years published for novel germplasm in accordance with the data management policy.	Complete	More than 100 datasets of detailing ranges of phenotypic, genotypic and genealogical data were released publicly in 2020 following data management policies and procedures.	https://data.cimmyt.org http://inside.cimmyt.org/datamanagement/SitePages/Breeding_Data_Management.aspx

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2020 extended to 2021 – (1) Lr67 knockouts identified in Reedling. (2) Screening for rust resistance initiated in greenhouse. (3) Guide RNA molecules designed for MLO, vectors made, and transformation initiated in Reedling.	Extended	Transformation of elite wheat lines has proven to be challenging, thus has presented a bottleneck in editing genes. Although we have generated more than two dozen transgenic plants with the editing machinery for the Lr67 gene in Reedling, none carried an edited copy of the gene. We have recently acquired new vectors from UC Davis that have been shown to improve both transformation and gene editing efficiencies. These vectors containing guide RNA molecules for Lr67 and MLO genes have just been transformed into Reedling. We expect to see results in 5-6 months and hope to identify edited copies for both the genes.	https://www.dropbox.com/work/WHEAT%20FP2%20Editing%20evidence?preview=Lr67+and+MLO++CRP+Milestone+2020.docx
2020 - Novel diversity available for yield potential, drought and heat tolerance in lines from crossing bank accessions with elite lines.	Complete	Materials derived from pre-breeding activities have been developed and distributed to CG and NARS partners. Examples include derivative lines from octoploid synthetic wheat lines (AABBDDDD and AABBDDAA), CIRNO derivatives developed for quality and leaf rust resistance and a suite of lines and trials from Seeds of Discovery incorporating novel diversity for drought, heat and disease tolerance.	https://www.dropbox.com/work/Maiz%20y%20Trigo%20para%20Mexico%202020%20Tech%20reports?preview=L04+Entregable+11+Trigo+2020+V290321.docx https://www.dropbox.com/sh/2v708sxz94h0lb/AABXRGpN6GSs2YMoQY3_n32aa?dl=0
2020 - Demonstrated use of genomic prediction to save phenotyping costs via 'sparse testing'.	Complete	Sparse testing implies modifying the original multi-environment breeding trials system (where all breeding germplasms are evaluated in all the testing environments) into a system where not all the breeding lines are sown in all environments. How best to establish a multi-environmental trial system that will be economically feasible, save land and water resources, without highly affecting the precision by which breeding lines or varieties' performance are assessed, predicted and selected? Sparse testing was applied to maize and wheat breeding of CIMMYT in order to study the genomic-enabled prediction accuracy of different field sparse testing systems. We studied three sparse testing allocation designs: (1) no-overlapping lines in environments, (2) complete overlapping (0 non-overlapping) lines tested in all of the environments and (3) combinations of the two previous cases where certain number of non-overlapping/overlapping lines were distributed in the environments. The study used three extensive wheat data sets and two large maize data sets. Results for sparse testing on wheat and maize data sets indicated that substantial save on testing	https://data.cimmyt.org/dataset.xhtml?persistentId=hdl:11529/10548552

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resources can be achieved by allocation designs and genome-based models. Reducing the size of the calibration sets always reduced the genomic-enabled prediction accuracy. The study manuscript is currently under peer review.

			2020 extended to 2021 - In collaboration with Julius Kuehn Institute and the Wheat Initiative in Berlin, develop global HeDWIC initiatives, including Doctoral Training Program.	Complete	The doctoral program was established with two enrolled PhD students in 2020. The global initiatives were developed in partnership with the Wheat Initiative. For the development of global initiatives HeDWIC and the Julius Kuehn Institute's TERTIUS project joined forces with AHEAD (an organization under the wheat initiative) to expand representation of wheat researchers working in heat and drought and to broaden the scope and scale of conceived initiatives.	https://hedwic.org/capacity-building/ https://www.wheatinitiative.org/ahead
			2020 extended to 2022 - Assess potential genetic gains if implementing Optimal Contributions Selection (which balances genetic gain with maintaining population diversity) in pre-breeding populations.	Extended	Improving parent selection in pre-breeding. The use of genome selection methods for pre-breeding will facilitate and improve the selection based on breeder knowledge, phenotypic data and genomic data. Differentiate the contribution of exotic and elite materials is of paramount importance to identify new sources of allele. Genome selection predictive models have been shown to be effective to do genome selection in exotic and pre-breeding materials. New genome selection predictive models contribute to have a better precision and facilitate the selection of crosses between exotic and elite materials.	http://digitalcommons.unl.edu/agronomyfacpub/890
FP2 Outcome: 2.5 Breeders develop improved varieties more efficiently through greater access and use of documented germplasm and tools	•Adoption of CGIAR materials with enhanced genetic gains	The refinement of a framework and development of resources to enable more efficient and effective storage and use of data for the purpose of varietal development has advanced comprehensively in 2020. This is in spite of a number of challenges related not only to the COVID-19 pandemic but also to a change in the structure and oversight of the EBS development. The work delivered moves WHEAT closer to comprehensive, efficient and effective breeding data management.	2020 extended to 2021 - Centralized breeding data management system and associated tools deployed to provide breeding teams with better access to germplasm, genealogical, and phenotypic data.	Extended	EBS v1.0 and v2.1 have been deployed to specific breeding team members at CIMMYT for the purpose of testing and training. Based on their specific feedback development sprints have been formulated and have and will be conducted to develop the next version for broader release to the MAIZE breeding community.	For limited distribution: https://ebs-staging-wheat.cimmyt.org/ https://www.youtube.com/watch?v=L_TVMYpYMdg https://ebsproject.atlassian.net/wiki/spaces/EBS/pages/10880646663/2020+EBS-B4R+Release

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2020 extended to 2022 - Centralized genotypic data management system deployed to provide breeding teams with better access to genotypic data	Extended	The GOBii project developed a centralized database for genotypic data. Breeding team members can access it though functionality is limited for direct breeding application. Breeders need to be able to select both genotypic data and phenotypic data for relevant germplasm entities (seed lots) to conduct needed analyses and select and advance germplasm. To facilitate this in a more integrated manner, the genotypic data management system is being incorporated into the EBS. While GOBii delivered a system the uplift into EBS is needed to enable broad use of data by breeding teams.	https://excellenceinbreeding.org/toolbox/tools/gobii-genomic-data-management-system-gobii-gdm http://gobii-pilot-02.cgiarad.org:8081/gobii-portal/
2020 - The effects of Mendelian sampling in cross performance was evaluated using phenotypic and genomic data from lines, crosses, and their progeny.	Changed	Improving parent selection for recycling of lines. The use of genome selection methods for parent selection will facilitate and improve the selection of parents based on breeder knowledge, phenotypic data and genomic data. The use of genome selection also allows a rapid recycling since phenotypic evaluation at early stages is more difficult. We contribute to developing analytical methods for breeders and scientists in order to incorporate genomic data in parent selection. New genome selection predictive models contribute to better precision and facilitate selection within the population. The new methods have performed well, however, more studies are necessary to incorporate environmental information more efficiently, to define the training population, and to complete the predictions in particular cases.	
2020 extended to 2021 - Centralized breeding data management system and associated tools deployed to provide breeders with better access to germplasm, genealogical, phenotypic, and genotypic data.	Extended	EBS v1.0 and v2.1 have been deployed to specific breeding team members at CIMMYT for the purpose of testing and training. Based on their specific feedback, development sprints have been formulated and will be conducted to develop the next version for broader release to the WHEAT breeding community.	For limited distribution: https://ebs-staging-wheat.cimmyt.org/ https://www.youtube.com/watch?v=L_TVMYpYMdg https://ebsproject.atlassian.net/wiki/spaces/EBS/pages/10880646663/2020+EBS-B4R+Release
2020 extended to 2021 - Centralized	Extended	EBS v1.0 and v2.1 have been deployed to specific breeding team members at CIMMYT for the purpose of	For limited distribution: https://ebs-staging-wheat.cimmyt.org/

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		breeding data management system and associated tools enhanced with prioritized functionalities for current and future users.		testing and training. Based on their specific feedback, development sprints have been formulated and will be conducted to develop the next version for broader release to the WHEAT breeding community.	https://www.youtube.com/watch?v=L_TVMYpYMdg https://ebsproject.atlassian.net/wiki/spaces/EBS/pages/10880646663/2020+EBS-B4R+Release
FP FP3 Outcome: 3.2 3 Partner breeding teams increased multidisciplinary and multi-institutional collaboration	•Enhance FP3 activities have been carried out with minimum disruption al Yield trials and nurseries have been composed and distributed of partner to the national programs on time. research organizational	2020 - Sustainable seed system optimized in 2-3 countries (pilots, with scaling-out potential).	Complete	Targeted partner countries, India, Nepal, Bangladesh Ethiopia and Nepal have engaged in pre-release and breeders seed multiplication of new varieties to ensure that sufficient seed is provided to seed multiplication agencies. These interventions and promotional activities increase the adoption of new varieties by farmers. The annual wheat improvement course was not carried out because of covid-19.	USAID Seed Project in Ethiopia (Bekele), DGGW project in Kenya and Ethiopia, BMZ project in India (Arun Joshi). Following 2020 publication using DNA fingerprint shows the evidence https://doi.org/10.1038/s41598-020-75181-8 . See also OICR4020 https://marlo.cgiar.org/projects/Wheat/study.do?expectedID=4020&edit=true&phaseID=152 and OICR4031.
		2020 - High yielding, disease-resistant varieties adopted by farmers, therefore crop losses reduced and less chemicals used in targeted countries.	Extended	All new varieties released by the partners are resistant to prevalent diseases- a prerequisite for the release in each of the target country. Use of fungicides to control diseases is nonexistent or remains minimal. Wheat yields in CIMMYT targeted countries in South Asia, West Asia, North Africa and SSA in the last 5 years increased at the rate of about 2% compared to a decade ago.	see Annual Reporting, sections 1.1 and 1.2 (FP3) and OICR4031: https://marlo.cgiar.org/projects/Wheat/study.do?expectedID=4031&edit=true&phaseID=152 See policies Table 2: P686, P694 see MELIA Table 10: S3961, S3962, S3967, S3973
FP3 Outcome: 3.3 Partner breeding teams improved breeding processes by adopting new technologies, methodologies, approaches and genetic resources	•Adoption of CGIAR materials with enhanced genetic gains	2020 – (1) More than 25 new, more productive, climate resilient, disease resistant varieties released by partner countries for enhancing wheat productivity and resilience. (2) More than 1000 improved wheat germplasm (varietal candidates) shared to	Complete	NA	see OICR4031: https://marlo.cgiar.org/projects/Wheat/study.do?expectedID=4031&edit=true&phaseID=152 see Annual Report, section 1.2., FP3International germplasm exchange for research maintained during COVID-19 times. CIMMYT and ICARDA invest \$700,000+ annually (mainly WHEAT W1&2) to distribute more than 350,000 packages of experimental seed (trials and nurseries) to over 220 IWIN collaborators worldwide. A majority of partners returned data voluntarily. Their key interest, according to a 2020 survey (87 respondents), was to obtain

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			more than 250 partners through various international trials and nurseries.			genetic diversity or parental germplasm (86.2%), or candidate varietal material (46%); most valuable to them was IWIN data on grain yield (76%) and diseases (75%). In addition, CIMMYT and ICARDA wheat genebanks fulfill seed requests from scientists globally on the basis of Standard Material Transfer Agreements (SMTA); this activity is funded by the Genebanks Platform.— See Table 4.
FP3 Outcome: 3.4	•Increase CRP commodities enhanced engagement in joint lobbying for speeding-up release of improved varieties	•Increased capacity for innovations in partner research organizations	2020 - Fully operational, integrated network of 15 precision phenotyping platforms, germplasm exchange between NARS platforms.	Extended	most field-based activities maintained, but training activities cancelled or virtualized, due to Covid-19	https://marlo.cgiar.org/projects/Wheat/description.do?projectId=1369&edit=true&phaseID=152
			2020 - Fully operational, integrated network of 6-8 precision phenotyping platforms developing and sharing information and germplasm with partners.	Complete		https://marlo.cgiar.org/projects/Wheat/description.do?projectId=1369&edit=true&phaseID=152
FP3 Outcome: 3.6	•Reduced National regulators of crop variety release improved enabling environment to speeding-up release of improved varieties	•Reduced smallholder producers' production risk	2020 - National variety release process 1-3 years shorter in 2-4 WHEAT target countries.	Complete	NA	https://marlo.cgiar.org/projects/Wheat/study.do?expectedID=4020&edit=true&phaseID=152 https://marlo.cgiar.org/projects/Wheat/study.do?expectedID=3929&edit=true&phaseID=152 https://marlo.cgiar.org/projects/Wheat/study.do?expectedID=3994&edit=true&phaseID=152 https://marlo.cgiar.org/projects/Wheat/study.do?expectedID=3962&edit=true&phaseID=152
			2020 - 4-5 WHEAT target countries develop culture of releasing new varieties	Complete	NA	https://marlo.cgiar.org/projects/Wheat/study.do?expectedID=3962&edit=true&phaseID=152 https://marlo.cgiar.org/projects/Wheat/policy.do?policyID=651&edit=true&phaseID=152

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			along with sufficient seed quantity.		https://marlo.cgiar.org/projects/Wheat/policy.do?policyID=694&edit=true&phaseID=152
FP3 Outcome: 3.7 Extension partners (universities, national/state/provincial governments) increased access and promotion of adoption of improved varieties to farmers, and increased investment in emerging private sector circumstances		2020 - Improved, documented understanding of specific wheat seed systems (farmer's seed commercial behavior, seed demand and marketing, economics of seed production) / 2-3 NARES-identified performance gaps, capacity development needs to identify and realize relevant capacity development interventions at apt levels.	Extended	Significant progress achieved in Morocco, India and Ethiopia, however much remains to be done to support NARS partners in sustaining change and building capacity.	https://marlo.cgiar.org/projects/Wheat/study.do?expectedID=3962&edit=true&phaseID=152 https://marlo.cgiar.org/projects/Wheat/policy.do?policyID=651&edit=true&phaseID=152 https://marlo.cgiar.org/projects/Wheat/policy.do?policyID=694&edit=true&phaseID=152
		2020 - Pre-release seed multiplication used in 3-4 countries to expedite seed availability to new varieties to seed producers.	Complete	NA	
FP3 Outcome: 3.12 Non-and-subsistence farmers adopted improved varieties	• Reduce pre- and post-harvest losses, including those caused by climate change	2020 extended to 2021 - Greater farmer adoption of released varieties (based on CGIAR research) in specific WHEAT target countries, compared to 1994-2014 average.	Extended	see OICRs 3929, 3994, 4031 and MELIA study Ethiopia wheat seed scaling.	Achieved. See India and Ethiopia case studies. https://marlo.cgiar.org/projects/Wheat/study.do?expectedID=3962&edit=true&phaseID=152 https://marlo.cgiar.org/projects/Wheat/policy.do?policyID=651&edit=true&phaseID=152 https://marlo.cgiar.org/projects/Wheat/policy.do?policyID=694&edit=true&phaseID=152 https://marlo.cgiar.org/projects/Wheat/study.do?expectedID=4020&edit=true&phaseID=152 This study estimated the genetic gains of the High Rainfall Trials (2006-2017) grown by NARS partners worldwide. The results indicate genetic gain ranging from 0.5-1%, per annum, which on average represents 46.6 kg/ha/year. The above described genetic gains impact ca. 61 million hectares on which farmers grow CGIAR-derived varieties, of 215 million global wheat area. https://doi.org/10.1016/j.fcr.2020.107742 https://marlo.cgiar.org/projects/Wheat/study.do?expectedID=4031&edit=true&phaseID=152

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		2020 - Superior varieties released by partner countries adopted by farmers to enhance productivity, climate resilience and reducing crop losses to diseases.	Complete	NA	see Annual Report, section 1.2 (FP3)
FP FP4 Outcome: 4.4	•				
4	NARS increased use of participatory approach in system research	Enhanced institutional capacity of partner research organizations	2020 - Series of training workshops conducted in South Asia and North Africa (with ICARDA)	Complete	INRA-Morocco funded wheat research continues, 2 proposal developments for PRIMA done, others ongoing.
					https://marlo.cgiar.org/projects/Wheat/description.do?projectID=1957&edit=true&phaseID=152 http://www.fao.org/3/ca9436en/ca9436en.pdf DOI: 10.3389/fpls.2020.00771.eCollection 2020
			2020 - CIMMYT and/or ICARDA have embedded farming systems and participatory approaches with NARS in a series of research projects in countries with interventions.	Complete	NA
					Nutrient Expert-based fertilizer management lowered global warming potential by 12-20% in wheat ; "MasAgro Productor partners continued to manage 38 innovation platforms in 32 Mexican states (see pp.6-7), in which 17,000 farmers engaged, adapting 36 agronomic practices. By 2020, the 449th extension agent graduated from the training program set up in 2009 - see MELIA Table, S3069 & S3997. Early sowing of wheat supports climate change adaptation. This package of new seed, machines and agronomy led to October sown wheat in 1 million hectares with about 0.8 t/ha extra yield and savings on irrigation. About ~0.3 million farmers benefitted with additional income (~US\$ 200/ha) and reduced environmental footprints - see SLO1.1, Table 1 https://marlo.cgiar.org/projects/Wheat/description.do?projectID=1957&edit=true&phaseID=152 http://www.fao.org/3/ca9436en/ca9436en.pdf doi: 10.3389/fpls.2020.00771.eCollection 2020
FP4 Outcome: 4.8	•	Technologies that reduce women's labor and energy expenditure	2020 - Increased adoption of combinations of SI strategies, technologies in specific target geographies compared to 2016.	Complete	INRA-Morocco funded wheat research continues, 2 proposal developments for PRIMA done, others ongoing.

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policies, processes re
and practices. adopted

		2020 - Adaptive research improves understanding of gender, youth and adoption, adaptation and scaling-up processes, with focus on market demand as trigger of innovation.	Complete	NA	Nutrient Expert-based fertilizer management lowered global warming potential by 12-20% in wheat ; MasAgro Productor partners continued to manage 38 innovation platforms in 32 Mexican states (see pp.6-7), in which 17,000 farmers engaged, adapting 36 agronomic practices. By 2020, the 449th extension agent graduated from the training program set up in 2009 - see MELIA Table, S3069 & S3997. Early sowing of wheat supports climate change adaptation. This package of new seed, machines and agronomy led to October sown wheat in 1 million hectares with about 0.8 t/ha extra yield and savings on irrigation. About ~0.3 million farmers benefitted with additional income (~US\$ 200/ha) and reduced environmental footprints - see SLO1.1, Table 1
FP4 Outcome: 4.9 Smallholder farmers increased their capacity to adopt and adapt SI e assets, practices and products (associated with crosscutting sub-IDO).	•Increase d access to productiv e assets, including natural resources	2020 - Farmers in two WHEAT (irrigated) target regions, using most water-efficient cultivars and optimum agronomy and irrigation systems, achieve water use efficiency of ca 450 l/kg grain can be achieved (50% water saving over 2015)	Complete	INRA-Morocco funded wheat research continues, 2 proposal developments for PRIMA done, others ongoing.	https://marlo.cgiar.org/projects/Wheat/description.do?projectId=1957&edit=true&phaseID=152 http://www.fao.org/3/ca9436en/ca9436en.pdf doi: 10.3389/fpls.2020.00771.eCollection 2020
		2020 – (1) Decision support-, mechanization and other tools, processes improve target groups' ability to seize opportunities and avoid losses. (2) Improved understanding of complex interaction	Complete	NA	Nutrient Expert-based fertilizer management lowered global warming potential by 12-20% in wheat , in addition to increased crop yield and incomes. The bottom-up analyses in India and Bangladesh suggest that zero tillage & residue management, as well as precision nutrient management could reduce country-level GHG emission by 17 and 17.5 million tonsCO₂e per year in India ; and by 0.55 and 7.98 million tons CO ₂ per year in Bangladesh.

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		between the enabling environment and business propositions for ICT services for scaling up, and social networking.			
FP4 Outcome: 4.6 Private sector (and public sector) increased provision of services to smallholder farmers to increase their ability to adopt SI practices and products	• Increase of beneficiaries to adopt research outputs	2020 - Series of last mile-collaborators actively involved in promotion and scaling of SI products. Reality check on scaling approaches implemented for development actors.	Complete	NA	Ex post, CCAFS/WHEAT: Assessment of factors affecting the adoption of Laser Land Leveling (LLL) and its impact on crop yields and net returns, based on HH survey data (621 randomly selected farmers, Haryana, India) and models. Adoption of LLL has significant positive impacts on yields (rice +549kg/ha; wheat +471kg/ha) and net returns (an aggregate increase of US\$230/ha) in the rice-wheat production system, raising farmers' income substantially. Link (OICR about mechanization)
FP4 Outcome: 4.10 Smallholder farmers adopted and adapted SI practices and products	• Closed yield gaps through improved agronomic and animal husbandry practices	2020 - Smart mechanization lessons learned routinely applied in other FP4 projects.	Complete	NA	Farm Power – Mechanization. Research has revealed much higher demand for mechanization in Eastern and Southern Africa, with African farming households being more dependent on labor markets than commonly assumed; and thus far more inclined to hire mechanization services . WHEAT and partner scientists compared case studies across 3 continents , reviewed Nepalese policy with a focus on appropriate-scale mechanization and investigated best-bet climate smart technologies, identifying the positive impact of appropriate mechanization on income and livelihoods; such as the Happy Seeder, which would offer farmers significant savings in production costs , amounting to US\$136, on top saving water and making timely wheat sowing easier. Approximately 11,000 Happy Seeders are in use in north-west India ; over 80% in Punjab. 18 manufacturers currently produce the Happy Seeder - see OICRs Table. Significant progress documented in 3 new OICRs. see SLO Table 1 & OICR Table 3.
		2020 - Continuous scaling up and out	Complete	NA	

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through W3 projects in all wheat regions.

2020 – (1) Increase resource-use efficiencies (irrigation water, N, P) while maintaining high, stable yields.	Complete	NA
(2) Extension of crop mgmt practices that arrest soil degradation.		
(3) Reduce labor burden and shortages.		

[robust nitrogen cycles in mid-hills and lowland \(Terai\) of Nepal](#). Nitrogen (N) is often the most limiting nutrient to productivity in smallholder mixed crop-livestock systems. The Terai farm N networks were organized (high productivity) but inflexible (poorly resilient) and consequently not sufficiently robust. Scenarios of improved management (improved seed, intercropping, use of fertilizers, better timing of activities) resulted in improved crop production. Farmer impacts and uptake of conservation agriculture. WHEAT, CCAFS and NARS partners [verified farmer benefits of adopting laser land levelling \(LLL\)](#): on yields (rice +549kg/ha–1; wheat +471kg/ha–1) and net returns (an aggregate increase of US\$230/ha) in the rice-wheat production system of Haryana, India. Science-based evidence, enhanced capacity and systems optimization such as zero-tillage and adapted varieties [led to advancing wheat seeding on more than 500,000 ha, benefiting over 800,000 smallholder farmers](#) in the eastern Indo-Gangetic Plains - see SLO 1.1, Table 1. In the Indo-Gangetic Plains of India, 0.87 million ha -- or 2.8% of total cultivated land -- is under zero-tillage, with multiple economic and environmental benefits. [ZT led to significant farmer cost savings](#), equal to a 5% increase in average total household incomes. [Nutrient Expert-based fertilizer management](#) lowered global warming potential by 12-20% in wheat, compared to conventional fertilization -- and increased crop yield and incomes - see SLO 1.2 & 3.2, Table 1.

Table 6: Numbers of peer-reviewed publications from current reporting period

(Sphere of control)

	Number	Percent
Peer-Reviewed publications	264	100.0%
Open Access	50	18.94%
ISI	59	22.35%

Table 7: Participants in CapDev Activities

Number of trainees	Female	Male
In short-term programs facilitated by CRP/PTF	2019	6462
In long-term programs facilitated by CRP/PTF	50	78
PhDs	10	19

Evidence Link: <https://wheat.org/download/cap-dev-data/>

Table 8: Key external partnerships

Lead FP	Brief description of partnership aims (30 words)	List of key partners in partnership. Do not use acronyms.	Main area of partnership (may choose multiple)
FP1	DNA (fingerprinting) extraction; survey analysis & policy engagement - see SLO Table 1.1	• EIAR - Ethiopian Institute of Agricultural Research	• Capacity Development • Research
FP1	Ex-ante impact assessment of biological nitrification inhibition (BNI) in wheat, sorghum, brachiaria	• JIRCAS - Japan International Research Center for Agricultural Sciences	• Research
FP2	Collaboration to study new synthetic wheat, amphiploids, and molecular information about wild species concerning climate adaptation/mitigation traits	• Tottori University • ALRC - Arid Land Research Center, Tottori University	• Research
FP3	Portable device MARPLE (Mobile and Real-time PLant disEase) detects which fungus pathogen strain is infecting a wheat crop in just 48 hours	• JIC - John Innes Centre	• Delivery
FP3	(G20) Wheat Initiative since 2012: Global data-sharing (Wheat Information System) & Expert Working Groups (EWG) participation	• G20WI - G20 Wheat Initiative	• Capacity Development
FP4	Systematic in-situ assessment of field-level energy-use and costs of groundwater irrigation in Nepal's Terai for preliminary evidence base to reduce groundwater access costs (P1764)	• The University of Manchester	• Research
FP4	Collaborative development of tools and systems for a climate-based wheat blast early warning system	• EMBRAPA - Empresa Brasileira de Pesquisa Agropecuária	• Research
FP4	New digital seed information system for all value chain actors, providing information on seed demand and supply in real time; under development	• NARC - Nepal Agricultural Research Council • SEAN - Seed Entrepreneurs' Association of Nepal	• Capacity Development • Delivery
FP4	Bring manufacturers of Agricultural Technology to new geographies in India. Linkage development and capacity-building with Department of Agriculture & Farmer Clubs (P884)	• Satmile SCOP	• Delivery • Capacity Development

Table 9: Internal Cross-CGIAR Collaborations

Brief description of the collaboration	Name(s) of collaborating CRP(s), Platform(s) or Center(s)	Optional: Value added, in a few words
Synergies re-enforcing the role of monitoring and big data	Big Data	Synergies
Synergies and collaborative work on foresight and targeting (PIM FP1); seed systems (PIM FP1); rural transformation (PIM FP2); markets/value chains (PIM FP3)	PIM	Synergies and collaborative work
Synergies and collaborative work on foresight and targeting (CoA1.1); adoption/impact (CoA1.2); gender/social inclusiveness (CoA1.3) and markets/value chains (CoA1.4)	Maize	Synergies and collaborative work
Synergies and collaborative work on gender/social inclusiveness; input into new platform	Gender	Synergies and collaborative work; scientific benefits; resource pooling
Joint foresight on how the cereal-based food security system has been evolving due to rural transformations . With an expected increase in population of 10.6 billion by 2050, the demand for cereals (including wheat) will rise to provide the base of global diets. The ability to supply more, high-quality, food critically depends on the capacity from rural areas; thus, it is imperative to understand rural transformation. This study indicates that the cereal-based food system will remain dominant.	Maize, PIM, Rice	Prioritization and targeting of R4D at CGIAR level

Table 10: Monitoring, Evaluation, Learning and Impact Assessment (MELIA)

Studies/learning exercises planned for this year (from POWB)	Status	Type of study or activity	Description of activity / study	Links to MELIA publications
S3057 - WHEAT Agri-food Systems & plant health-related synthesis/learning products of foresight & targeted research	Ongoing	Other MELIA activity	By linking crop models with economic models and approaches, crop model outputs can be used as inputs into socioeconomic modeling efforts for priority setting and policy advice. This review covers global modeling efforts, using the IMPACT model to create farm household bio-economic models and assess the potential impact of new technologies on farming systems and livelihoods. The value of crop model outputs--and underpinning crop science knowledge and data--is greatly enhanced by adding the socioeconomic context. There is a clear distinction between foresight and ex-ante impact assessment. Amongst the wide variety of topics that can be addressed with bio-economic modeling approaches, climate change is arguably the most noteworthy.	https://doi.org/10.1002/csc2.20114
S3061 - Collaborative analysis of gender norms and wheat agriculture in patriarchal contexts	Completed	Synthesis (secondary) study	This study builds on other recent studies investigating (enabling environments for) women's empowerment. Social norms that encapsulate men as farmers and women as homemakers' conflict with the reality that women are workers and decision makers in wheat production systems (Farnworth et al. 2020). Changes in perceptions and norms around farming actors would potentially change the dynamics of household decision making, leading to adoption of new agricultural technologies.	https://doi.org/10.1057/s41287-020-00281-0 https://doi.org/10.1080/02255189.2019.1650332
S3067 - International expert workshop on improving CIMMYT breeding methodologies	Completed	Other MELIA activity	The AG2MW international expert panel workshop participants (Mexico, Jan 28-30, 2020) discussed key considerations for successful changes in the CIMMYT Wheat breeding program, changes to the breeding scheme and the future role of genomic selection/prediction. Conclusions and recommendations flowed into detailed AGG project design and work planning. Internal documentation only.	https://www.cimmyt.org/news/cimmyt-scientists-join-fellow-experts-in-san-diego-for-worlds-largest-plant-and-animal-genomics-conference/

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S3068 - Synthesis and lessons learned from innovation hub approaches in South Asia, including their scaling potential	Ongoing	Synthesis (secondary) study	Developing and documenting key learnings about scaling pathways for Conservation Agriculture for Sustainable Intensification technologies and methods in South Asian countries, based on multiple programs and projects' experiences (e.g. csisa.org, bisa.org), as well as re-thinking adoption dynamics (e.g. technological change as a complex sociotechnical reconfiguration, situated in time and space). Ongoing learning process.	https://www.cimmyt.org/publications/new-publication-scaling-agricultural-mechanization-services-in-smallholder-farming-systems/ https://www.cimmyt.org/news/are-solar-powered-irrigation-systems-scalable/ https://doi.org/10.1177%2F0030727019864978
S3069 - Refine country collaboration methodology, approach	Completed	Other MELIA activity	This project sought to further develop and document the Integrated Agri-food Systems Initiative (IASI) approach, based on Maize for Mexico and Maize for Columbia stakeholder engagement experiences. Building on diverse existing data resources and leveraging situation analysis, modeled predictions, and scenarios, this holistic, multi-sector methodology synchronizes public and private action at the national level toward sustainable, equitable and inclusive agri-food systems. This culminates in collectively agreed strategies and multi-partner tactical plans, fosters mindset shifts and stakeholder consensus on sustainable and scalable innovations that respond to real dynamics in complex agri-food systems. Paper will be published in 2021.	https://repository.cimmyt.org/handle/10883/20219 https://repository.cimmyt.org/handle/10883/20218
S3070 - Multi-criteria approaches for better targeting and enhanced adoption of technologies	Ongoing	Other MELIA activity	This study examines prospects of the smallholder production systems in South Asia, SSA & other regions to adapt to climatic variability to minimize the negative impacts of climate change on food systems; understand macro- and micro-level factors that drive change in farming systems and communities; and understand why farmers use few adaptation measures, in light of the existing barriers and policy setup. Institutional change and flexibility to enable adaptation are a key challenge; clear need to further develop FARMDesign model and methodologies to assist in designing and implementing change at farm level.	https://doi.org/10.1007/s10668-019-00414-4 https://doi.org/10.1038/s41893-020-0500-2 https://doi.org/10.1016/j.agsy.2019.102695 https://doi.org/10.1080/20421338.2019.1694780 https://doi.org/10.1016/j.agsy.2020.102942
S3071 - Multi-partner scaling conference and CGIAR Scaling Expert Network meeting	Ongoing	Other MELIA activity	Bring together several international networks working on scaling approaches, including Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ)-funded CGIAR Scaling Experts network. Postponed into 2021 due to Covid19 crisis. To increase impact opportunities from projects, topics of discussion and workshops will consider (i) Exchange lessons on scaling positive impacts of innovations and programs around the world; (ii) Discuss ways of mainstreaming scaling in donor, research and development organizations; (iii) Reach a common understanding of	https://www.cimmyt.org/events/the-art-and-science-of-scaling-impact-for-smallholder-agriculture/ https://www.cimmyt.org/work/sustainable-intensification/scaling/

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			the status of the art and science of scaling; (iv) Strengthen the network of experts in scaling, agriculture and rural development.	
S3961 - First nationally representative, large-scale wheat DNA fingerprinting study	Completed	Ex-post adoption study	This study undertook a plot-level comparison of DNA fingerprinting results with farmer recall from nearly 4000 plots in the 2016/17 season. Results indicate that only 28% of farmers correctly named wheat varieties grown. New, rust resistant varieties are now widely adopted. 94% of the plots and 90% of the area sampled contained CGIAR-derived varieties. Scientists established the extent of crop varietal misidentification: correct identification was positively associated with farmer education and seed purchases from trusted sources (cooperatives, known farmers). Combining household survey and DNA fingerprinting supports reliable varietal adoption, impact assessments, and generates useful knowledge to inform varietal replacement and seed systems development.	https://doi.org/10.1371/journal.pone.0235484 https://doi.org/10.1038/s41598-020-75181-8
S3962 - Lessons Learned from the Wheat Seed Scaling Project 2015-2019	Completed	Program/project adoption or impact assessment	Lessons Learned Report, Wheat Seed Scaling Project in Ethiopia, 2015-2019 (CIMMYT, EIAR; funded by USAID) documents important lessons learned and recommendations (pp. 105-6) concerning the fast-track variety release process, which relies on W1&2-funded international germplasm exchange and NARS partnerships (see Chapter 3) and the un-sustained success of establishing a durum value chain for national pasta production (Chapter 7).	https://repository.cimmyt.org/bitstream/handle/10883/20837/61703.pdf?sequence=4&isAllowed=y
S3964 - Prospects of S Asian smallholder production systems to adapt to climatic variability, minimizing negative CC impacts on food systems	Completed	Correlates of adoption/impact study	WHEAT and CCAFS scientists examined the how South Asian smallholder production systems might adapt to climatic variability to minimize the negative impacts of climate change on food systems. Scientists looked at why farmers use few adaptation measures, if any, despite their prevalence, in light of the existing barriers and policy set-up. Institutional change and flexibility to enable adaptation are key challenges, as the adaptive management approaches are new to the existing bureaucratic systems in these counties. Education and interaction among farmers changed their adaptation behavior. Focusing on farmer-to-farmer communication can help to improve adaptation.	https://doi.org/10.1007/s10668-019-00414-4
S3966 - Crop residue burning, air pollution and mitigation evidence	Completed	Ex-ante, baseline and/or foresight study	Evidence gathering on profitability for farmers to adopt no-burn technologies. Every year after the rice harvest, some 2.5 million farmers in northwest India burn the remaining stubble to prepare their fields for the subsequent wheat crop, causing massive air pollution affecting millions. WHEAT scientists examined different tillage practices as empirical evidence including 'Happy Seeder' (HS). There is no yield penalty; savings in wheat production costs (136 USD ha ⁻¹) &	https://doi.org/10.1080/14735903.2020.1834277

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for different tillage practices			water use; facilitates timely sowing. Private benefits of HS use combined with its societal benefits justify dedicated policy support for scaling; plus, stricter enforcement of banning residue burning.	
S3967 - Understanding low level of adoption of new durum wheat cultivars in Morocco	Completed	Program/project adoption or impact assessment	WHEAT scientists wanted to understand if the low level of adoption of new durum wheat cultivars in Morocco can be partially explained by a misalignment of breeders' objectives. Factorial analysis was applied to data obtained through interviews with 90 wheat breeders from 49 countries and a case study of 861 durum wheat growers in Morocco, revealing the existence of five breeder and four farmer classes. The results indicate that a certain degree of misalignment exists between farmers' and breeders' objectives, and that the use of participatory farmer-weighted indexes could help raise the level of adoption.	https://doi.org/10.20900/cbagg20200014
S3968 - Conservation agriculture-based wheat production system to minimize losses due to climate risks	Completed	Correlates of adoption/impact study	This multi-year farmer adoption study & on-farm experiment, CCAFS/WHEAT: conservation agriculture-based wheat production systems (CAW) can serve as ex-ante measure to minimize loss due to climate risks (e.g. extreme rainfall) in India; CAW reduces GHG emissions, air pollution; yield during a bad year was 4.89 Mg/ha under CAW, which is higher than the conventional control (4.23 Mg/ha). Most farmers who had applied CAW continued to practice it and increased the proportion of land area allocated. Removing barriers to adoption can help enhance farmers' income with and without climatic extremes. The authors proposed 4 improvement measures.	http://doi.org/10.1108/IJCCSM-09-2018-0065
S3970 - Designing profitable, resource-use efficient, environmentally sound cereal-based systems for the Western Indo-Gangetic plains	Completed	Ex-ante, baseline and/or foresight study	This ex-ante, 4-year on-station field study substituted Conservation Agriculture-based maize for rice and found that similar mean profitability gains were realized (16%) and transformative improvements in irrigation (– 84%), energy use efficiency (+ 231%), and global warming potential (– 95%) were observed, compared to the Farmers' Practice (FP) from before. Including mung beans in rotation further increased system productivity, profitability, and energy use efficiency; decreasing irrigation water use by 64% and global warming potential by 106%, compared to FP. Despite the described considerable benefits, adoption of maize is not widespread due to uneven market demand and assured price guarantees for rice.	https://doi.org/10.1038/s41598-020-76035-z
S3971 - Assessing technological change in agri-food	Completed		This adoption-impact study analyzed farmer acceptance of innovations and their impacts: what was taken up, how it works in farmers' fields and what are the	https://journals.sagepub.com/doi/10.1177/030727020930728

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systems of the Global South: Meta-analysis of adoption-impact studies in wheat systems		Synthesis (secondary) study	implications thereof. This study reviews 47 empirical studies (2008–2017). Their coverage and scope does not fully capture the complex, dynamic technological changes taking place, for they portray adoption as binary choice between old and new (superior). Most studies look at short-term changes (e.g. change in yield, profitability), often assuming that improved living standards will follow; labor productivity, ecological and social impacts are largely ignored. The authors propose to enhance data availability, analytical rigor, and reorient future research toward socially, economically, ecologically relevant dimensions.	
S3972 - Estimating adoption and impacts of farm/agronomic practices in developing countries using satellite data	Completed	Synthesis (secondary) study	A scoping review of 54 studies found that: (1) satellite data was successfully used to detect agricultural practices (e.g. cropping intensity, soil conservation); (2) few studies estimated yield impacts, although estimating crop yields with satellite data is fairly developed; and (3) few studies explored impact estimation beyond the biophysical sphere. Estimation of certain environmental impacts of agricultural practices is possible, but rarely done. Few studies assessed the economic impacts of interventions. Satellite data analysis allows information access with little delay and over longer periods, over wide geographies, whilst reducing measurement error in certain variables. More interdisciplinary research needed to foster uptake.	https://doi.org/10.1007/s13593-020-0610-2
S3973 - Meta-analysis of small-scale producers' adoption of climate-resilient crops in the Global South over the last 30 years	Completed	Synthesis (secondary) study	A scoping review with MAIZE/WHEAT scientist contributions examined conditions leading to climate-resilient crop adoption in countries experiencing climate-related impacts (e.g. 45 indicators) and the predominance of cereals in adoption studies (67%). Small-scale producers needed to cope with abiotic stresses (drought, heat and flooding). The most prevalent traits were drought tolerance and water-use efficiency. Adoption was driven by 1) availability and effectiveness of extension services, 2) education levels, 3) farmers' access to inputs—especially seeds and fertilizers, and 4) socio-economic status. When farmers adopt multiple complementary strategies, they help to build agriculture systems that can respond to shocks. A key recommendation of the study (out of 8): Go for a multiple-interventions approach when promoting climate-resilient crops adoption.	
S3976 - Conservation Agriculture-based practices: Major	Completed	Ex-ante, baseline and/or	Recent studies confirmed significant potential for conservation agriculture (CA) suitability and adoption by farmers, including for Morocco: 260,000 ha (33% of total cereal area). However, no-tillage surface in Morocco stands at 21,000 ha in	https://repo.mel.cgiar.org/handle/20.500.11766/10943 https://doi.org/10.1016/j.scitotenv.2019.07.307

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potential to improve economic, biophysical viability and sustainability of MENA region dryland agriculture		foresight study	2020 (Mrabet et al. 2021). MENA-wide, farmer CA practice covers less than 5% of total cereal area, raising questions about suitability. Along with benefits for integrated crop-livestock systems, WHEAT and partner scientists showed that CA has major potential to improve the economic and biophysical viability and sustainability. Several factors explain low adoption, including: lack of affordable and adapted seeders; ill-conceived policies inadvertently promoting monocropping; other.	https://dx.doi.org/10.3390/su12093640 https://dx.doi.org/10.1016/j.agsy.2020.102946
S3978 - Review of Nepalese agricultural mechanization policy since 2014	Completed	Program/project evaluation/review	Agricultural mechanization in developing countries has taken at least two contested innovation pathways—the “incumbent trajectory” that promotes industrial agriculture, and an “alternative pathway” that supports small-scale mechanization. Authors examine the case of Nepal’s first Agricultural Mechanization Promotion Policy (AMPP) since 2014 using a conceptual framework of what will be defined as “responsible innovation”. They make 3 recommendations (consider smallholder land fragmentation; reflexivity and innovation champions; domestic machinery production).	https://doi.org/10.3390/su12010374
S3979 - Gender-sensitive agricultural mechanization policy and programs in Nepal	Completed	Qualitative outcome study: (mainly to substantiate contribution to policy or similar)	Scientists show there are substantial differences in the adoption of small-scale farm mechanization between male-headed (MH-HHs) and female-headed households (FH-HHs). These differences are larger when compared with food-secure and food-insecure FH-HHs. The prospect of adopting mini-tillers for female-headed households would remain lower even after having the same level of attributes for the MH-HHs and FH-HHs.	https://doi.org/10.1016/j.techsoc.2020.101250
S3980 - International Wheat Yield Partnership Hub (IWYP Hub) at CIMMYT under WHEAT FP2: Progress report	Ongoing	Other MELIA activity	The IWYP Hub serves as the central technical platform for evaluation of the innovations generated and a development pipeline for building them into elite adapted germplasm for distribution connects fundamental research through CGIAR-led applied research institutes targeting the Global South. IWYP Hub research/pre-breeding with Wheat Physiology team included: (i)Explored genetic diversity for (I) HI and identify improved selection approaches, (ii) traits related to canopy photosynthesis. (ii)Genome-wide association study (GWAS) analysis from experimental population data sets. (iii)Pre-breeding for yield potential and delivery of Wheat Yield Collaboration Yield Trials (WYCYT) nurseries including application of genomic selection. (iv)Marker-assisted selection for HI.	https://www.sciencedirect.com/science/article/pii/S0168945219315699?via%3Dihub https://onlinelibrary.wiley.com/doi/full/10.1002/fes3.241 https://ieeexplore.ieee.org/document/9130745 https://www.frontiersin.org/articles/10.3389/fpls.2020.549743/full https://access.onlinelibrary.wiley.com/doi/abs/10.1002/csc2.20098 https://www.sciencedirect.com/science/article/pii/S0378429019313632?via%3Dihub https://www.sciencedirect.com/science/article/abs/pii/S037842902031217X

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				https://www.sciencedirect.com/science/article/pii/S0378429019313632 https://www.mdpi.com/2073-4395/10/10/1527 https://academic.oup.com/jxb/article/71/7/2299/5576008 https://onlinelibrary.wiley.com/doi/full/10.1111/pbr.12885 https://www.dropbox.com/h?preview=Annual+Report+Format+IWYP+Hub.docx
S3987 - Morocco's seed system makes progress, but some challenges still remain	Completed	Qualitative Outcome Study: (mainly to substantiate contribution to policy or similar)	A 2013 survey of the seed system value chain documented that despite tremendous national and international efforts, varietal turn-over was still very low (e.g. varieties released within the last 20 years were grown on only 42% of wheat area). The "freedom to act" in the seed business resulted in the creation of over 300 new seed enterprises with more regional foci, initiating a shift in certified seed sales to favor NON-expired European Union plant variety protection (PVP), enabling significant varietal turn-over. INRA-CGIAR-derived varieties (mostly pre-1990) lost market share (29% in 2019). The "Boudour" project invested US\$45 million in certified seed sales.	https://hdl.handle.net/20.500.11766/12701 https://doi.org/10.20900/cbagg20200014 http://dx.doi.org/20.500.11766/8505
S3997 - Heat and Drought Wheat Improvement Consortium (HEDWIC) at CIMMYT: progress report	Completed	Other MELIA activity	HeDWIC activities and accomplishments in 2020 include: Raising \$5 million in new funds for the HeDWIC translational research and breeding Hub in Mexico, through a proposal submitted in March to FFAR titled "Harnessing translational research across a global wheat improvement network for climate resilience. HeDWIC collaborated in 5 other proposals submitted in 2020 on heat tolerance, leveraging the HeDWIC Hub project research activities. HeDWIC formed a strategic advisory board (SAB) that met twice with activity leaders to review the goals and objectives, holding the first meeting in December.	https://www.sciencedirect.com/science/article/pii/S0168945219315699 https://link.springer.com/article/10.1007/s00122-019-03454-6 https://www.sciencedirect.com/science/article/pii/S0168945219315699 https://www.nature.com/articles/s43016-019-0004-2 https://www.dropbox.com/h?preview=2020+HeDWIC+Outputs+(002).docx https://hedwic.org/
S4015 - Usefulness of breeder-friendly phenotyping approaches and methods at	Completed	Ex-ante, baseline and/or foresight study	This review considers the 'breeder friendliness' of phenotyping within three domains: 1) the 'minimum data set', where it is paramount that data is easy to collect and use; (2) high throughput phenotyping (HTP); (3) detailed characterization or 'precision' phenotyping, typically customized for a set of traits associated with a target environment. Extra investment for phenotyping is becoming more accepted to capitalize on recent developments in crop genomics	https://doi.org/10.1016/j.plantsci.2019.110396 https://hdl.handle.net/10883/20776

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different stages of the breeding cycle			and prediction models. This review considers different contexts for phenotyping, including breeding, exploration of genetic resources, parent building and translational research, and how the different categories of phenotyping apply to each.	
S4016 - Review of model-based approaches (modelling) to assist breeding activities under climate change (focus on CGIAR breeding)	Completed	Ex-ante, baseline and/or foresight study	This review addresses the question: 'To what extent can breeding gains be achieved under a changing climate, at a pace sufficient to usefully contribute to climate adaptation, mitigation and food security?' As CGIAR breeding programs are undergoing a major modernization process, crop modelers will need to be part of crop improvement teams, with a common understanding of breeding pipelines and model capabilities/limitations, common data standards and protocols, to ensure they follow and deliver according to clearly-defined breeding products. This will enable more rapid and better-targeted crop modeling activities, thus directly contributing to accelerated and more impactful breeding efforts.	https://doi.org/10.1002/csc2.20048
S4021 - Pakistan: Speeding up the national trials and variety release process for wheat	On Going	Program/project adoption or impact assessment	In Pakistan, the long-term USDA Wheat Productivity Enhancement Program (WPEP) supported national and local wheat research institutes technically and financially in releasing better varieties faster; resulting in the release of more than 40 high yielding, disease resistant varieties since 2010. Generation advancement and testing candidate lines has thus kept pace with the rapidly evolving rust pathogens and erratic environmental changes.	https://www.wpepforpakistan.org/Uploads/News/Documents/iwrqca11.2n1.pdf
S4033 - MAIZE, RICE, PIM, WHEAT joint foresight: Cereal-based food security system has been evolving due to rural transformations	Completed	Ex-ante, baseline and/or foresight study	The cereal-based food security system has been evolving due to rural transformations. As the ability to supply more food, particularly high-quality foods, critically depends on the capacity of food supply from rural areas, it is imperative to understand rural transformation. This study indicates that a cereal-based food system will remain dominant, so the challenges are 1) to support rural and agricultural transformation (e.g. poverty alleviation, employment creation) while maintaining agricultural diversity; and 2) for major cereals to meet the increased demand without displacing other nutritious crops and livestock from rural landscapes.	https://doi.org/10.1016/j.gfs.2020.100441

Table 11: Update on Actions Taken in Response to Relevant Evaluations

Name of the evaluation	Recommendation number (from evaluation)	Text of recommendation (can be shortened)	Status of response to this recommendation	Concrete actions taken for this recommendation.	By whom (per action)	When (per action)	Link to evidence
NA							

NA

Table 12: Examples of W1/2 Use in this reporting period (2020)

Please give specific examples, one per row (including through set aside strategic research funds or partner funds)

Select broad area of use of W1/2 from the categories below - (drop down) Select only one category.

WHEAT is guided by the high-level framework for W1&2 deployment shown below, while Table 12 (pp.18-19 above) shows in more detail where W1&2 has been invested during 2020, based on [83 work packages in the W1&2-per-FP annual work plan](#) delivered CIMMYT and ICARDA scientists and their implementation partners.

Table 13: CRP Financial Report

CRP ON WHEAT - CRP17

Reported Period - 2020

Expenses by Flagships	Planned Budget				Actual Expenditures				Difference			
	Windows 1 & 2	Window 3	Bilateral	Total	Windows 1 & 2	Window 3	Bilateral	Total	Windows 1 & 2	Window 3	Bilateral	Total
FP1 Inclusive and profitable wheat opportunities	1,865	3,273	705	5,843	1,411	2,010	376	3,797	454	1,263	329	2,046
FP2 Novel diversity and tools for improving genetic gains and breeding efficiency	3,240	2,008	2,648	7,896	2,546	1,052	2,363	5,961	694	956	285	1,935
FP3 Better varieties reach farmers faster	5,518	5,378	6,622	17,518	3,954	3,262	4,556	11,772	1,564	2,116	2,066	5,746
FP4 Sustainable intensification of wheat-based farming systems	2,238	8,034	3,182	13,454	1,978	5,303	2,803	10,084	260	2,731	379	3,370
Capacity Development				-	-	-	-	-	-	-	-	-
Strategic Competitive Research grant				-	-	-	-	-	-	-	-	-
Management and Support Cost	1,230	-	-	1,230	1,076	-	-	1,076	154	-	-	154
Total Costs	14,091	18,693	13,157	45,941	10,965	11,627	10,098	32,690	3,126	7,066	3,059	13,251
Note: CGIAR Collaboration Costs breakdown by FP												
FP1 Inclusive and profitable wheat opportunities	255			255	217	-	-	217	38	-	-	38
FP2 Novel diversity and tools for improving genetic gains and breeding efficiency	607			607	592	-	-	592	15	-	-	15
FP3 Better varieties reach farmers faster	1,309			1,309	1,175	-	-	1,175	134	-	-	134
FP4 Sustainable intensification of wheat-based farming systems	391			391	334	784	57	1,175	57	(784)	(57)	(784)
Strategic Competitive Research grant				-	-	-	-	-	-	-	-	-
Management and Support Cost	130			130	170	-	-	170	(40)	-	-	(40)
Total CGIAR Collaboration Costs	2,692	-	-	2,692	2,488	784	57	3,329	204	(784)	(57)	(637)
TOTAL CRP ON WHEAT (Net w/o CGIAR Collaborations)	11,399	18,693	13,157	43,249	8,477	10,843	10,041	29,361	2,922	7,850	3,116	13,888



**RESEARCH
PROGRAM ON
Wheat**

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